122 $T\tau$ AFM collinear spin layer groups

Néel directions, SOC magnetic layer groups and layer-resolved intrinsic nonlinear Hall constraints

Coverage. All 122 entries of Table S4, in the original order, expanded into **738 direction branches**: 2 groups have 1 branch, 38 have 3, 70 have 7, and 12 have 11. Directions related by a parent symmetry are retained in the fixed source coordinate setting so that the displayed tensor indices stay meaningful. These 738 rows are direction strata, not 738 distinct MLG types.
Source. J.-X. Du, S. Zeng and Y.-J. Zhao, *Stacking theory for bilayer two-dimensional magnets*, Phys. Rev. B **113**, 214455 (2026), DOI: 10.1103/g471-xrkl; Supplemental Material, Table S4, pp. 34–63. The source supplies the cSLGs and operations. The Néel-direction branches, SOC MLGs and response constraints in this report are calculated results.
Direction notation. $\mathrm{AX}_{[uvw]}$ denotes a special axis; $\mathrm{PL}_{(hkl)}$ denotes a direction in a special plane excluding all axes lying in that plane; GEN denotes a direction outside every listed axis and plane. Only symmetry-distinct stabilizer strata of a given group are introduced; for example, groups 1.3.3 and 2.4.7 have a single GEN row covering all directions.
Coordinates and numbering. z is normal to the physical layers; Cartesian x, y follow the rectangular or tetragonal source family-cell axes $\mathbf{a}, \mathbf{b}$. Direction indices refer to that source cell, not to an enlarged magnetic cell. Table S4 uses **OG notation**; the complete lattice subscripts, symbols and three-part identifiers are retained. The 122 nonrelativistic $T\tau$ AFM groups are distinct from the 66 groups called type-IV two-dimensional magnets in Table S6.
Operation columns. Translation parts are omitted and repeated point operations are merged. All distinct nontrivial-part SLG point operations are shown on separate lines; the common spin-only factor is implicit. Every MLG row explicitly lists its point operations on one line. Full MLG symbols and numbers are nevertheless identified from the full affine operations in the source, then verified by a basis/origin change preserving the layer normal. MLG symbols use the reference OG settings; the explicitly listed operations and tensor indices remain in the source coordinate setting. An affine witness for every row is included in the data package. Omitting translations from the displayed columns does not redefine these group identifiers. In these projected columns, $\{-1\|E\}$ and $\mathcal{T}$ originate from anti-translations; they do not assert bare time-reversal symmetry of the full spatial group. The magnetic translation lattice is retained in every identification and affine check.
Tensor convention. Set $A^\ell = \chi^\ell_{xyy}$ and $B^\ell = \chi^\ell_{yxx}$ for $\ell = t, b$, with $\chi_{abc} = -\chi_{bac}$ and in-plane indices. Then $\chi_{xyx} = -B$, $\chi_{yxy} = -A$, and all other in-plane entries vanish. The two-component vector $\mathbf{v}^\ell = (A^\ell, B^\ell)$ transforms as $\mathbf{v}^{g\ell} = q_g R_\parallel \mathbf{v}^\ell$. This statement is specific to the $\mathcal{T}$-odd intrinsic Hall part.

Mark	Constraint on the displayed component	Meaning
$+$	$\chi^t = \chi^b$, allowed	Equal layer contributions
$-$	$\chi^t = -\chi^b$, allowed	Hidden, layer compensated
0	$\chi^t = \chi^b = 0$	Forbidden in each layer
v	No fixed proportional relation between the same component in the two layers	See coupled constraints when starred

Uniform result. Every listed SOC subgroup contains $\{\mathcal{T} \mid \boldsymbol{\tau}\}$ with $\tau_z = 0$. This operation preserves each physical layer and reverses the $\mathcal{T}$-odd, spatially averaged intrinsic Hall tensor. Consequently $\chi^t_{xyy} = \chi^b_{xyy} = \chi^t_{yxx} = \chi^b_{yxx} = 0$ for all **738 branches**. The two response columns therefore contain 0 throughout. These are layer-resolved zeros; no locally finite layer contribution survives to form a hidden interlayer pair. Appendix A gives the derivation and magnetic-cell conventions.
Assumptions and calculation. Fixed structure and collinear order; two physical layers at opposite sides of $z = 0$. The SOC subgroup satisfies $q_g \det(R_g) R_g \hat{\mathbf{n}} = s_g \hat{\mathbf{n}}$, with $q_g = +1$ for g and $q_g = -1$ for $g\mathcal{T}$. Operations with $R_{zz} = -1$ exchange the two layers. Branches are constructed from all real ± 1 eigenspaces and their intersections. These are symmetry-allowed constrained orientations, not predictions of magnetic easy axes or stability.
Source audit (original entries retained in the raw data). Two isolated translation entries violate multiplication by the explicitly listed anti-translation $(1, 0, 0)$:

S1. 50.4.361 (p. 56), operation 06: use $(1, 0, 0)$ for $\{-1\|C_{2z}\}$, replacing the printed $(1/2, 0, 0)$.

S2. 61.11.432 (p. 61), operation 25: use $(1, 0, 0)$ for $\{-1\|\mathcal{P}\}$, replacing the printed $(0, 0, 0)$.

Both repairs follow directly by multiplying operation 05 or 17 (the anti-translation) by operation 02 or 09, respectively. All 122 corrected lists close in their magnetic translation lattices and agree with the printed SLG symbols. These are consistency repairs, not author-issued errata; they do not change the projected operations or the universal zero-response result.
Validation. All 738 MLG assignments are checked against complete affine operations. Each allowed response space is independently checked using the full third-rank Hall tensor, and direction coverage is sampled on every axis and plane as well as generic orientations.

Complete table, in Table S4 order

First three columns are merged across all direction branches of one cSLG. S1–S2 refer to the source audit. All response entries are zero by the layer-preserving anti-translation constraint.

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG (OG) symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
01 1.3.3 $p^1_{2a}1$ SI p. 34	$T\tau$ AFM	$\{1\|E\}$ $\{-1\|E\}$	GEN	$p_{2a}1$ 1.3.3	$\{E,\ \mathcal{T}\}$	0	0
02 2.4.7 $p^1_{2a}\bar{1}$ SI p. 34	$T\tau$ AFM	$\{1\|E\}$ $\{1\|\mathcal{P}\}$ $\{-1\|E\}$ $\{-1\|\mathcal{P}\}$	GEN	$p_{2a}\bar{1}$ 2.4.7	$\{E,\ \mathcal{P},\ \mathcal{T},\ \mathcal{PT}\}$	0	0
03 3.4.11 $p^1_{2a}1^11^12$ SI p. 34	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|E\}$ $\{-1\|C_{2z}\}$	$\mathrm{AX}_{[001]}$	$p_{2a}112$ 3.4.11	$\{E,\ C_{2z},\ \mathcal{T},\ C_{2z}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}\overline{112}$ 3.4.11	$\{E,\ C_{2z}\mathcal{T},\ \mathcal{T},\ C_{2z}\}$	0	0
			GEN	$p_{2a}\bar{1}$ 1.3.3	$\{E,\ \mathcal{T}\}$	0	0
04 4.4.15 $p^1_{2a}1^11^1m$ SI p. 34	$T\tau$ AFM	$\{1\|E\}$ $\{1\|M_z\}$ $\{-1\|E\}$ $\{-1\|M_z\}$	$\mathrm{AX}_{[001]}$	$p_{2a}11m$ 4.4.15	$\{E,\ M_z,\ \mathcal{T},\ M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}11m'$ 4.5.16	$\{E,\ M_z\mathcal{T},\ \mathcal{T},\ M_z\}$	0	0
			GEN	$p_{2a}\bar{1}$ 1.3.3	$\{E,\ \mathcal{T}\}$	0	0
05 4.5.16 $p^1_{2a}1^11^{-1}m$ SI p. 34	$T\tau$ AFM	$\{1\|E\}$ $\{-1\|M_z\}$ $\{-1\|E\}$ $\{1\|M_z\}$	$\mathrm{AX}_{[001]}$	$p_{2a}11m'$ 4.5.16	$\{E,\ M_z\mathcal{T},\ \mathcal{T},\ M_z\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}11m$ 4.4.15	$\{E,\ M_z,\ \mathcal{T},\ M_z\mathcal{T}\}$	0	0
			GEN	$p_{2a}\bar{1}$ 1.3.3	$\{E,\ \mathcal{T}\}$	0	0
06 5.4.20 $p^1_{2b}1^11^1a$ SI p. 34	$T\tau$ AFM	$\{1\|E\}$ $\{1\|M_z\}$ $\{-1\|E\}$ $\{-1\|M_z\}$	$\mathrm{AX}_{[001]}$	$p_{2b}11a$ 5.4.20	$\{E,\ M_z,\ \mathcal{T},\ M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2b}\overline{11}a$ 5.4.20	$\{E,\ M_z\mathcal{T},\ \mathcal{T},\ M_z\}$	0	0
			GEN	$p_{2a}\bar{1}$ 1.3.3	$\{E,\ \mathcal{T}\}$	0	0
07 6.6.26 $p^1_{2a}1^11^12/^1m$ SI p. 34	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{-1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_z\}$	$\mathrm{AX}_{[001]}$	$p_{2a}112/m$ 6.6.26	$\{E,\ C_{2z},\ \mathcal{P},\ M_z,\ \mathcal{T},\ C_{2z}\mathcal{T},\ \mathcal{PT},\ M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}112'/m'$ 6.7.27	$\{E,\ C_{2z}\mathcal{T},\ \mathcal{P},\ M_z\mathcal{T},\ \mathcal{T},\ C_{2z},\ \mathcal{PT},\ M_z\}$	0	0
			GEN	$p_{2a}\bar{1}$ 2.4.7	$\{E,\ \mathcal{P},\ \mathcal{T},\ \mathcal{PT}\}$	0	0

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG (OG) symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
08 6.7.27 $p^1_{2a}1^11^{-1}2/^{-1}m$ SI p. 34	$T\tau$ AFM	$\{1\|E\}$ $\{-1\|C_{2z}\}$ $\{1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{-1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|\mathcal{P}\}$ $\{1\|M_z\}$	$\mathrm{AX}_{[001]}$	$p_{2a}112'/m'$ 6.7.27	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, \mathcal{T}, C_{2z}, \mathcal{PT}, M_z\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}112/m$ 6.6.26	$\{E, C_{2z}, \mathcal{P}, M_z, \mathcal{T}, C_{2z}\mathcal{T}, \mathcal{PT}, M_z\mathcal{T}\}$	0	0
			GEN	$p_{2a}\bar{1}$ 2.4.7	$\{E, \mathcal{P}, \mathcal{T}, \mathcal{PT}\}$	0	0
09 7.6.33 $p^1_{2b}1^11^12/^1a$ SI p. 34	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{-1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_z\}$	$\mathrm{AX}_{[001]}$	$p_{2b}112/a$ 7.6.33	$\{E, C_{2z}, \mathcal{P}, M_z, \mathcal{T}, C_{2z}\mathcal{T}, \mathcal{PT}, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2b}112/a$ 7.6.33	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, \mathcal{T}, C_{2z}, \mathcal{PT}, M_z\}$	0	0
			GEN	$p_{2a}\bar{1}$ 2.4.7	$\{E, \mathcal{P}, \mathcal{T}, \mathcal{PT}\}$	0	0
10 8.4.37 $p^{-1}_{2a}2^11^11$ SI p. 35	$T\tau$ AFM	$\{1\|E\}$ $\{-1\|C_{2x}\}$ $\{-1\|E\}$ $\{1\|C_{2x}\}$	$\mathrm{AX}_{[100]}$	$p_{2a}2'11$ 8.4.37	$\{E, C_{2x}\mathcal{T}, \mathcal{T}, C_{2x}\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_{2a}211$ 8.6.39	$\{E, C_{2x}, \mathcal{T}, C_{2x}\mathcal{T}\}$	0	0
			GEN	$p_{2a}\bar{1}$ 1.3.3	$\{E, \mathcal{T}\}$	0	0
11 8.5.38 $p^1_c2^11^11$ SI p. 35	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2x}\}$ $\{-1\|E\}$ $\{-1\|C_{2x}\}$	$\mathrm{AX}_{[100]}$	p_c211 8.5.38	$\{E, C_{2x}, \mathcal{T}, C_{2x}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(100)}$	p_c211 8.5.38	$\{E, C_{2x}\mathcal{T}, \mathcal{T}, C_{2x}\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0
12 8.6.39 $p^1_{2a}2^11^11$ SI p. 35	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2x}\}$ $\{-1\|E\}$ $\{-1\|C_{2x}\}$	$\mathrm{AX}_{[100]}$	$p_{2a}211$ 8.6.39	$\{E, C_{2x}, \mathcal{T}, C_{2x}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_{2a}2'11$ 8.4.37	$\{E, C_{2x}\mathcal{T}, \mathcal{T}, C_{2x}\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0
13 8.7.40 $p^1_{2b}2^11^11$ SI p. 35	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2x}\}$ $\{-1\|E\}$ $\{-1\|C_{2x}\}$	$\mathrm{AX}_{[100]}$	$p_{2b}211$ 8.7.40	$\{E, C_{2x}, \mathcal{T}, C_{2x}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_{2b}211$ 8.7.40	$\{E, C_{2x}\mathcal{T}, \mathcal{T}, C_{2x}\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0
14 9.4.44 $p^1_{2b}2^1_11^11$ SI p. 35	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2x}\}$ $\{-1\|E\}$ $\{-1\|C_{2x}\}$	$\mathrm{AX}_{[100]}$	$p_{2b}2_111$ 9.4.44	$\{E, C_{2x}, \mathcal{T}, C_{2x}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_{2b}2_111$ 9.4.44	$\{E, C_{2x}\mathcal{T}, \mathcal{T}, C_{2x}\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG (OG) symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
15 10.4.48 $c_p^1 2^1 1^1 1$ SI p. 35	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2x}\}$ $\{-1\|E\}$ $\{-1\|C_{2x}\}$	$AX_{[100]}$	$c_p 211$ 10.4.48	$\{E, C_{2x}, \mathcal{T}, C_{2x}\mathcal{T}\}$	0	0
			$PL_{(100)}$	$c_p 2'11$ 10.5.49	$\{E, C_{2x}\mathcal{T}, \mathcal{T}, C_{2x}\}$	0	0
			GEN	$p_{2a}\overline{1}$ 1.3.3	$\{E, \mathcal{T}\}$	0	0
16 10.5.49 $c_p^{-1} 2^1 1^1 1$ SI p. 35	$T\tau$ AFM	$\{1\|E\}$ $\{-1\|C_{2x}\}$ $\{-1\|E\}$ $\{1\|C_{2x}\}$	$AX_{[100]}$	$c_p 2'11$ 10.5.49	$\{E, C_{2x}\mathcal{T}, \mathcal{T}, C_{2x}\}$	0	0
			$PL_{(100)}$	$c_p 211$ 10.4.48	$\{E, C_{2x}, \mathcal{T}, C_{2x}\mathcal{T}\}$	0	0
			GEN	$p_{2a}\overline{1}$ 1.3.3	$\{E, \mathcal{T}\}$	0	0
17 11.4.53 $p_{2a}^1 m^1 1^1 1$ SI p. 35	$T\tau$ AFM	$\{1\|E\}$ $\{1\|M_x\}$ $\{-1\|E\}$ $\{-1\|M_x\}$	$AX_{[100]}$	$p_{2a}m11$ 11.4.53	$\{E, M_x, \mathcal{T}, M_x\mathcal{T}\}$	0	0
			$PL_{(100)}$	$p_{2a}m11$ 11.4.53	$\{E, M_x\mathcal{T}, \mathcal{T}, M_x\}$	0	0
			GEN	$p_{2a}\overline{1}$ 1.3.3	$\{E, \mathcal{T}\}$	0	0
18 11.5.54 $p_{2b}^1 m^1 1^1 1$ SI p. 35	$T\tau$ AFM	$\{1\|E\}$ $\{1\|M_x\}$ $\{-1\|E\}$ $\{-1\|M_x\}$	$AX_{[100]}$	$p_{2b}m11$ 11.5.54	$\{E, M_x, \mathcal{T}, M_x\mathcal{T}\}$	0	0
			$PL_{(100)}$	$p_{2b}m'11$ 11.6.55	$\{E, M_x\mathcal{T}, \mathcal{T}, M_x\}$	0	0
			GEN	$p_{2a}\overline{1}$ 1.3.3	$\{E, \mathcal{T}\}$	0	0
19 11.6.55 $p_{2b}^{-1} m^1 1^1 1$ SI p. 35	$T\tau$ AFM	$\{1\|E\}$ $\{-1\|M_x\}$ $\{-1\|E\}$ $\{1\|M_x\}$	$AX_{[100]}$	$p_{2b}m'11$ 11.6.55	$\{E, M_x\mathcal{T}, \mathcal{T}, M_x\}$	0	0
			$PL_{(100)}$	$p_{2b}m11$ 11.5.54	$\{E, M_x, \mathcal{T}, M_x\mathcal{T}\}$	0	0
			GEN	$p_{2a}\overline{1}$ 1.3.3	$\{E, \mathcal{T}\}$	0	0
20 11.7.56 $p_c^1 m^1 1^1 1$ SI p. 36	$T\tau$ AFM	$\{1\|E\}$ $\{1\|M_x\}$ $\{-1\|E\}$ $\{-1\|M_x\}$	$AX_{[100]}$	$p_c m11$ 11.7.56	$\{E, M_x, \mathcal{T}, M_x\mathcal{T}\}$	0	0
			$PL_{(100)}$	$p_c m11$ 11.7.56	$\{E, M_x\mathcal{T}, \mathcal{T}, M_x\}$	0	0
			GEN	$p_{2a}\overline{1}$ 1.3.3	$\{E, \mathcal{T}\}$	0	0
21 12.4.60 $p_{2a}^1 b^1 1^1 1$ SI p. 36	$T\tau$ AFM	$\{1\|E\}$ $\{1\|M_x\}$ $\{-1\|E\}$ $\{-1\|M_x\}$	$AX_{[100]}$	$p_{2a}b11$ 12.4.60	$\{E, M_x, \mathcal{T}, M_x\mathcal{T}\}$	0	0
			$PL_{(100)}$	$p_{2a}b11$ 12.4.60	$\{E, M_x\mathcal{T}, \mathcal{T}, M_x\}$	0	0
			GEN	$p_{2a}\overline{1}$ 1.3.3	$\{E, \mathcal{T}\}$	0	0
22 13.4.64 $c_p^1 m^1 1^1 1$ SI p. 36	$T\tau$ AFM	$\{1\|E\}$ $\{1\|M_x\}$ $\{-1\|E\}$ $\{-1\|M_x\}$	$AX_{[100]}$	$c_p m11$ 13.4.64	$\{E, M_x, \mathcal{T}, M_x\mathcal{T}\}$	0	0
			$PL_{(100)}$	$c_p m'11$ 13.5.65	$\{E, M_x\mathcal{T}, \mathcal{T}, M_x\}$	0	0
			GEN	$p_{2a}\overline{1}$ 1.3.3	$\{E, \mathcal{T}\}$	0	0

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG (OG) symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
23 13.5.65 $c_p^{-1}m^11^11$ SI p. 36	$T\tau$ AFM	$\{1\|E\}$ $\{-1\|M_x\}$ $\{-1\|E\}$ $\{1\|M_x\}$	$AX_{[100]}$	$c_pm'11$ 13.5.65	$\{E, M_x\mathcal{T}, \mathcal{T}, M_x\}$	0	0
			$PL_{(100)}$	c_pm11 13.4.64	$\{E, M_x, \mathcal{T}, M_x\mathcal{T}\}$	0	0
			GEN	$p_{2a}\bar{1}$ 1.3.3	$\{E, \mathcal{T}\}$	0	0
24 14.6.71 $p_{2a}^{-1}2/^{-1}m^11^11$ SI p. 36	$T\tau$ AFM	$\{1\|E\}$ $\{-1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{-1\|M_x\}$ $\{-1\|E\}$ $\{1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{1\|M_x\}$	$AX_{[100]}$	$p_{2a}2'/m'11$ 14.6.71	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}, \mathcal{T}, C_{2x}, \mathcal{PT}, M_x\}$	0	0
			$PL_{(100)}$	$p_{2a}2/m11$ 14.9.74	$\{E, C_{2x}, \mathcal{P}, M_x, \mathcal{T}, C_{2x}\mathcal{T}, \mathcal{PT}, M_x\mathcal{T}\}$	0	0
			GEN	$p_{2a}\bar{1}$ 2.4.7	$\{E, \mathcal{P}, \mathcal{T}, \mathcal{PT}\}$	0	0
25 14.7.72 $p_{2b}^{-1}2/^{-1}m^11^11$ SI p. 36	$T\tau$ AFM	$\{1\|E\}$ $\{-1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{-1\|M_x\}$ $\{-1\|E\}$ $\{1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{1\|M_x\}$	$AX_{[100]}$	$p_{2b}2'/m'11$ 14.7.72	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}, \mathcal{T}, C_{2x}, \mathcal{PT}, M_x\}$	0	0
			$PL_{(100)}$	$p_{2b}2/m11$ 14.10.75	$\{E, C_{2x}, \mathcal{P}, M_x, \mathcal{T}, C_{2x}\mathcal{T}, \mathcal{PT}, M_x\mathcal{T}\}$	0	0
			GEN	$p_{2a}\bar{1}$ 2.4.7	$\{E, \mathcal{P}, \mathcal{T}, \mathcal{PT}\}$	0	0
26 14.8.73 $p_c^12/^1m^11^11$ SI p. 36	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_x\}$ $\{-1\|E\}$ $\{-1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_x\}$	$AX_{[100]}$	$p_c2/m11$ 14.8.73	$\{E, C_{2x}, \mathcal{P}, M_x, \mathcal{T}, C_{2x}\mathcal{T}, \mathcal{PT}, M_x\mathcal{T}\}$	0	0
			$PL_{(100)}$	$p_c2/m11$ 14.8.73	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}, \mathcal{T}, C_{2x}, \mathcal{PT}, M_x\}$	0	0
			GEN	$p_{2a}\bar{1}$ 2.4.7	$\{E, \mathcal{P}, \mathcal{T}, \mathcal{PT}\}$	0	0
27 14.9.74 $p_{2a}^12/^1m^11^11$ SI p. 36	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_x\}$ $\{-1\|E\}$ $\{-1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_x\}$	$AX_{[100]}$	$p_{2a}2/m11$ 14.9.74	$\{E, C_{2x}, \mathcal{P}, M_x, \mathcal{T}, C_{2x}\mathcal{T}, \mathcal{PT}, M_x\mathcal{T}\}$	0	0
			$PL_{(100)}$	$p_{2a}2'/m'11$ 14.6.71	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}, \mathcal{T}, C_{2x}, \mathcal{PT}, M_x\}$	0	0
			GEN	$p_{2a}\bar{1}$ 2.4.7	$\{E, \mathcal{P}, \mathcal{T}, \mathcal{PT}\}$	0	0

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG (OG) symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
28 14.10.75 $p^{1}_{2b}2/^{1}m^{1}1^{1}1$ SI p. 37	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_x\}$ $\{-1\|E\}$ $\{-1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$p_{2b}2/m11$ 14.10.75	$\{E,\, C_{2x},\, \mathcal{P},\, M_x,\, \mathcal{T},\, C_{2x}\mathcal{T},\, \mathcal{P}\mathcal{T},\, M_x\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_{2b}2'/m'11$ 14.7.72	$\{E,\, C_{2x}\mathcal{T},\, \mathcal{P},\, M_x\mathcal{T},\, \mathcal{T},\, C_{2x},\, \mathcal{P}\mathcal{T},\, M_x\}$	0	0
			GEN	$p_{2a}\bar{1}$ 2.4.7	$\{E,\, \mathcal{P},\, \mathcal{T},\, \mathcal{P}\mathcal{T}\}$	0	0
29 15.6.81 $p^{-1}_{2b}2_1/^{-1}m^{1}1^{1}1$ SI p. 37	$T\tau$ AFM	$\{1\|E\}$ $\{-1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{-1\|M_x\}$ $\{-1\|E\}$ $\{1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$p_{2b}2'_1/m'11$ 15.6.81	$\{E,\, C_{2x}\mathcal{T},\, \mathcal{P},\, M_x\mathcal{T},\, \mathcal{T},\, C_{2x},\, \mathcal{P}\mathcal{T},\, M_x\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_{2b}2_1/m11$ 15.7.82	$\{E,\, C_{2x},\, \mathcal{P},\, M_x,\, \mathcal{T},\, C_{2x}\mathcal{T},\, \mathcal{P}\mathcal{T},\, M_x\mathcal{T}\}$	0	0
			GEN	$p_{2a}\bar{1}$ 2.4.7	$\{E,\, \mathcal{P},\, \mathcal{T},\, \mathcal{P}\mathcal{T}\}$	0	0
30 15.7.82 $p^{1}_{2b}2_1/^{1}m^{1}1^{1}1$ SI p. 37	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_x\}$ $\{-1\|E\}$ $\{-1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$p_{2b}2_1/m11$ 15.7.82	$\{E,\, C_{2x},\, \mathcal{P},\, M_x,\, \mathcal{T},\, C_{2x}\mathcal{T},\, \mathcal{P}\mathcal{T},\, M_x\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_{2b}2'_1/m'11$ 15.6.81	$\{E,\, C_{2x}\mathcal{T},\, \mathcal{P},\, M_x\mathcal{T},\, \mathcal{T},\, C_{2x},\, \mathcal{P}\mathcal{T},\, M_x\}$	0	0
			GEN	$p_{2a}\bar{1}$ 2.4.7	$\{E,\, \mathcal{P},\, \mathcal{T},\, \mathcal{P}\mathcal{T}\}$	0	0
31 16.6.88 $p^{-1}_{2a}2/^{-1}b^{1}1^{1}1$ SI p. 37	$T\tau$ AFM	$\{1\|E\}$ $\{-1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{-1\|M_x\}$ $\{-1\|E\}$ $\{1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$p_{2a}2'/b'11$ 16.6.88	$\{E,\, C_{2x}\mathcal{T},\, \mathcal{P},\, M_x\mathcal{T},\, \mathcal{T},\, C_{2x},\, \mathcal{P}\mathcal{T},\, M_x\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_{2a}2/b11$ 16.7.89	$\{E,\, C_{2x},\, \mathcal{P},\, M_x,\, \mathcal{T},\, C_{2x}\mathcal{T},\, \mathcal{P}\mathcal{T},\, M_x\mathcal{T}\}$	0	0
			GEN	$p_{2a}\bar{1}$ 2.4.7	$\{E,\, \mathcal{P},\, \mathcal{T},\, \mathcal{P}\mathcal{T}\}$	0	0
32 16.7.89 $p^{1}_{2a}2/^{1}b^{1}1^{1}1$ SI p. 37	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_x\}$ $\{-1\|E\}$ $\{-1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$p_{2a}2/b11$ 16.7.89	$\{E,\, C_{2x},\, \mathcal{P},\, M_x,\, \mathcal{T},\, C_{2x}\mathcal{T},\, \mathcal{P}\mathcal{T},\, M_x\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_{2a}2'/b'11$ 16.6.88	$\{E,\, C_{2x}\mathcal{T},\, \mathcal{P},\, M_x\mathcal{T},\, \mathcal{T},\, C_{2x},\, \mathcal{P}\mathcal{T},\, M_x\}$	0	0
			GEN	$p_{2a}\bar{1}$ 2.4.7	$\{E,\, \mathcal{P},\, \mathcal{T},\, \mathcal{P}\mathcal{T}\}$	0	0

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG (OG) symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
33 18.6.100 $c^1_p2/^1m^11^11$ SI p. 38	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_x\}$ $\{-1\|E\}$ $\{-1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_x\}$	$AX_{[100]}$	$c_p2/m11$ 18.6.100	$\{E, C_{2x}, \mathcal{P}, M_x, \mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$PL_{(100)}$	$c_p2'/m'11$ 18.8.102	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}, \mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_x\}$	0	0
			GEN	$p_{2a}\bar{1}$ 2.4.7	$\{E, \mathcal{P}, \mathcal{T}, \mathcal{P}\mathcal{T}\}$	0	0
34 18.7.101 $c^1_p2/^{-1}m^11^11$ SI p. 38	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_x\}$ $\{-1\|E\}$ $\{-1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_x\}$	$AX_{[100]}$	$c_p2/m'11$ 18.7.101	$\{E, C_{2x}, \mathcal{P}\mathcal{T}, M_x\mathcal{T}, \mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}, M_x\}$	0	0
			$PL_{(100)}$	$c_p2'/m11$ 18.9.103	$\{E, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_x, \mathcal{T}, C_{2x}, \mathcal{P}, M_x\mathcal{T}\}$	0	0
			GEN	$p_{2a}\bar{1}$ 2.4.7	$\{E, \mathcal{P}\mathcal{T}, \mathcal{T}, \mathcal{P}\}$	0	0
35 18.8.102 $c_p^{-1}2/^{-1}m^11^11$ SI p. 38	$T\tau$ AFM	$\{1\|E\}$ $\{-1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{-1\|M_x\}$ $\{-1\|E\}$ $\{1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{1\|M_x\}$	$AX_{[100]}$	$c_p2'/m'11$ 18.8.102	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}, \mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_x\}$	0	0
			$PL_{(100)}$	$c_p2/m11$ 18.6.100	$\{E, C_{2x}, \mathcal{P}, M_x, \mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_x\mathcal{T}\}$	0	0
			GEN	$p_{2a}\bar{1}$ 2.4.7	$\{E, \mathcal{P}, \mathcal{T}, \mathcal{P}\mathcal{T}\}$	0	0
36 18.9.103 $c_p^{-1}2/^1m^11^11$ SI p. 38	$T\tau$ AFM	$\{1\|E\}$ $\{-1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{1\|M_x\}$ $\{-1\|E\}$ $\{1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{-1\|M_x\}$	$AX_{[100]}$	$c_p2'/m11$ 18.9.103	$\{E, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_x, \mathcal{T}, C_{2x}, \mathcal{P}, M_x\mathcal{T}\}$	0	0
			$PL_{(100)}$	$c_p2/m'11$ 18.7.101	$\{E, C_{2x}, \mathcal{P}\mathcal{T}, M_x\mathcal{T}, \mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}, M_x\}$	0	0
			GEN	$p_{2a}\bar{1}$ 2.4.7	$\{E, \mathcal{P}\mathcal{T}, \mathcal{T}, \mathcal{P}\}$	0	0

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG (OG) symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
37 19.5.108 $p^{-1}_{2a}2^{-1}2^12$ SI p. 38	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{-1\|E\}$ $\{-1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$	$\mathrm{AX}_{[100]}$	$p_{2a}2'2'2$ 19.5.108	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{T}, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\}$	0	0
			$\mathrm{AX}_{[010]}$	$p_{2a}222$ 19.7.110	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{T}, C_{2z}, C_{2y}, C_{2x}\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_{2a}222$ 19.7.110	$\{E, C_{2z}, C_{2y}, C_{2x}, \mathcal{T}, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}112$ 3.4.11	$\{E, C_{2z}\mathcal{T}, \mathcal{T}, C_{2z}\}$	0	0
			$\mathrm{PL}_{(010)}$	$p_{2b}211$ 8.7.40	$\{E, C_{2y}, \mathcal{T}, C_{2y}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_{2a}211$ 8.6.39	$\{E, C_{2x}, \mathcal{T}, C_{2x}\mathcal{T}\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0
38 19.6.109 $p^1_c2^12^12$ SI p. 38	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{-1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$	$\mathrm{AX}_{[100]}$	p_c222 19.6.109	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{T}, C_{2z}, C_{2y}, C_{2x}\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	p_c222 19.6.109	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{T}, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\}$	0	0
			$\mathrm{AX}_{[001]}$	p_c222 19.6.109	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{T}, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}112$ 3.4.11	$\{E, C_{2z}\mathcal{T}, \mathcal{T}, C_{2z}\}$	0	0
			$\mathrm{PL}_{(010)}$	p_c211 8.5.38	$\{E, C_{2y}\mathcal{T}, \mathcal{T}, C_{2y}\}$	0	0
			$\mathrm{PL}_{(100)}$	p_c211 8.5.38	$\{E, C_{2x}\mathcal{T}, \mathcal{T}, C_{2x}\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0
39 19.7.110 $p^1_{2a}2^12^12$ SI p. 39	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{-1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$	$\mathrm{AX}_{[100]}$	$p_{2a}222$ 19.7.110	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{T}, C_{2z}, C_{2y}, C_{2x}\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	$p_{2a}2'2'2$ 19.5.108	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{T}, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_{2a}2'2'2$ 19.5.108	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{T}, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}112$ 3.4.11	$\{E, C_{2z}\mathcal{T}, \mathcal{T}, C_{2z}\}$	0	0
			$\mathrm{PL}_{(010)}$	$p_{2b}211$ 8.7.40	$\{E, C_{2y}\mathcal{T}, \mathcal{T}, C_{2y}\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_{2a}2'11$ 8.4.37	$\{E, C_{2x}\mathcal{T}, \mathcal{T}, C_{2x}\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0
40 20.6.116 $p^{-1}_{2b}2^{-1}_12^12$ SI p. 39	$T\tau$ AFM	$\{1\|E\}$ $\{-1\|C_{2x}\}$ $\{-1\|C_{2y}\}$ $\{1\|C_{2z}\}$ $\{-1\|E\}$ $\{1\|C_{2x}\}$ $\{1\|C_{2y}\}$ $\{-1\|C_{2z}\}$	$\mathrm{AX}_{[100]}$	$p_{2b}2_122$ 20.7.117	$\{E, C_{2x}\mathcal{T}, C_{2y}, C_{2z}\mathcal{T}, \mathcal{T}, C_{2x}, C_{2y}\mathcal{T}, C_{2z}\}$	0	0
			$\mathrm{AX}_{[010]}$	$p_{2b}2'_12'2$ 20.6.116	$\{E, C_{2x}, C_{2y}\mathcal{T}, C_{2z}\mathcal{T}, \mathcal{T}, C_{2x}\mathcal{T}, C_{2y}, C_{2z}\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_{2b}2_122$ 20.7.117	$\{E, C_{2x}, C_{2y}, C_{2z}, \mathcal{T}, C_{2x}\mathcal{T}, C_{2y}\mathcal{T}, C_{2z}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}112$ 3.4.11	$\{E, C_{2z}\mathcal{T}, \mathcal{T}, C_{2z}\}$	0	0
			$\mathrm{PL}_{(010)}$	$p_{2a}211$ 8.6.39	$\{E, C_{2y}, \mathcal{T}, C_{2y}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_{2b}2_111$ 9.4.44	$\{E, C_{2x}, \mathcal{T}, C_{2x}\mathcal{T}\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG (OG) symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
41 20.7.117 $p^1_{2b}2^1_12^12$ SI p. 39	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2x}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2z}\}$ $\{-1\|E\}$ $\{-1\|C_{2x}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2z}\}$	$\mathrm{AX}_{[100]}$	$p_{2b}2'_12'2$ 20.6.116	$\{E, C_{2x}, C_{2y}\mathcal{T}, C_{2z}\mathcal{T}, \mathcal{T}, C_{2x}\mathcal{T}, C_{2y}, C_{2z}\}$	0	0
			$\mathrm{AX}_{[010]}$	$p_{2b}2_122$ 20.7.117	$\{E, C_{2x}\mathcal{T}, C_{2y}, C_{2z}\mathcal{T}, \mathcal{T}, C_{2x}, C_{2y}\mathcal{T}, C_{2z}\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_{2b}2'_12'2$ 20.6.116	$\{E, C_{2x}\mathcal{T}, C_{2y}\mathcal{T}, C_{2z}, \mathcal{T}, C_{2x}, C_{2y}, C_{2z}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}112$ 3.4.11	$\{E, C_{2z}\mathcal{T}, \mathcal{T}, C_{2z}\}$	0	0
			$\mathrm{PL}_{(010)}$	$p_{2a}2'11$ 8.4.37	$\{E, C_{2y}\mathcal{T}, \mathcal{T}, C_{2y}\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_{2b}2_111$ 9.4.44	$\{E, C_{2x}\mathcal{T}, \mathcal{T}, C_{2x}\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0
42 22.5.126 $c^1_p2^12^12$ SI p. 39	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{-1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$	$\mathrm{AX}_{[100]}$	$c_p2'22'$ 22.7.128	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{T}, C_{2z}, C_{2y}, C_{2x}\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	$c_p2'22'$ 22.7.128	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{T}, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\}$	0	0
			$\mathrm{AX}_{[001]}$	$c_p2'2'2$ 22.6.127	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{T}, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}112$ 3.4.11	$\{E, C_{2z}\mathcal{T}, \mathcal{T}, C_{2z}\}$	0	0
			$\mathrm{PL}_{(010)}$	$c_p2'11$ 10.5.49	$\{E, C_{2y}\mathcal{T}, \mathcal{T}, C_{2y}\}$	0	0
			$\mathrm{PL}_{(100)}$	$c_p2'11$ 10.5.49	$\{E, C_{2x}\mathcal{T}, \mathcal{T}, C_{2x}\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0
43 22.6.127 $c^{-1}_p2^{-1}2^12$ SI p. 39	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{-1\|E\}$ $\{-1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$	$\mathrm{AX}_{[100]}$	$c_p2'22'$ 22.7.128	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{T}, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\}$	0	0
			$\mathrm{AX}_{[010]}$	$c_p2'22'$ 22.7.128	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{T}, C_{2z}, C_{2y}, C_{2x}\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	c_p222 22.5.126	$\{E, C_{2z}, C_{2y}, C_{2x}, \mathcal{T}, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}112$ 3.4.11	$\{E, C_{2z}\mathcal{T}, \mathcal{T}, C_{2z}\}$	0	0
			$\mathrm{PL}_{(010)}$	c_p211 10.4.48	$\{E, C_{2y}, \mathcal{T}, C_{2y}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(100)}$	c_p211 10.4.48	$\{E, C_{2x}, \mathcal{T}, C_{2x}\mathcal{T}\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0
44 22.7.128 $c^{-1}_p2^12^{-1}2$ SI p. 40	$T\tau$ AFM	$\{1\|E\}$ $\{-1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{-1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{1\|C_{2x}\}$	$\mathrm{AX}_{[100]}$	$c_p2'2'2$ 22.6.127	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{T}, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\}$	0	0
			$\mathrm{AX}_{[010]}$	c_p222 22.5.126	$\{E, C_{2z}, C_{2y}, C_{2x}, \mathcal{T}, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$c_p2'22'$ 22.7.128	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{T}, C_{2z}, C_{2y}, C_{2x}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}112$ 3.4.11	$\{E, C_{2z}, \mathcal{T}, C_{2z}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	$c_p2'11$ 10.5.49	$\{E, C_{2y}\mathcal{T}, \mathcal{T}, C_{2y}\}$	0	0
			$\mathrm{PL}_{(100)}$	c_p211 10.4.48	$\{E, C_{2x}, \mathcal{T}, C_{2x}\mathcal{T}\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG (OG) symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
45 23.5.133 $p^{-1}_{2a}m^{-1}m^{1}2$ SI p. 40	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$ $\{-1\|E\}$ $\{-1\|C_{2z}\}$ $\{1\|M_y\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$p_{2a}mm2$ 23.7.135	$\{E, C_{2z}\mathcal{T}, M_y, M_x\mathcal{T}, \mathcal{T}, C_{2z}, M_y\mathcal{T}, M_x\}$	0	0
			$\mathrm{AX}_{[010]}$	$p_{2a}m'm'2$ 23.5.133	$\{E, C_{2z}\mathcal{T}, M_y\mathcal{T}, M_x, \mathcal{T}, C_{2z}, M_y, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_{2a}mm2$ 23.7.135	$\{E, C_{2z}, M_y, M_x, \mathcal{T}, C_{2z}\mathcal{T}, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}112$ 3.4.11	$\{E, C_{2z}\mathcal{T}, \mathcal{T}, C_{2z}\}$	0	0
			$\mathrm{PL}_{(010)}$	$p_{2b}m11$ 11.5.54	$\{E, M_y, \mathcal{T}, M_y\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_{2a}m11$ 11.4.53	$\{E, M_x, \mathcal{T}, M_x\mathcal{T}\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0
46 23.6.134 $p^{1}_{c}m^{1}m^{1}2$ SI p. 40	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|M_y\}$ $\{1\|M_x\}$ $\{-1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$p_{c}mm2$ 23.6.134	$\{E, C_{2z}\mathcal{T}, M_y\mathcal{T}, M_x, \mathcal{T}, C_{2z}, M_y, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	$p_{c}mm2$ 23.6.134	$\{E, C_{2z}\mathcal{T}, M_y, M_x\mathcal{T}, \mathcal{T}, C_{2z}, M_y\mathcal{T}, M_x\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_{c}mm2$ 23.6.134	$\{E, C_{2z}, M_y\mathcal{T}, M_x\mathcal{T}, \mathcal{T}, C_{2z}\mathcal{T}, M_y, M_x\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}112$ 3.4.11	$\{E, C_{2z}\mathcal{T}, \mathcal{T}, C_{2z}\}$	0	0
			$\mathrm{PL}_{(010)}$	$p_{c}m11$ 11.7.56	$\{E, M_y\mathcal{T}, \mathcal{T}, M_y\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_{c}m11$ 11.7.56	$\{E, M_x\mathcal{T}, \mathcal{T}, M_x\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0
47 23.7.135 $p^{1}_{2a}m^{1}m^{1}2$ SI p. 40	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|M_y\}$ $\{1\|M_x\}$ $\{-1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$p_{2a}m'm'2$ 23.5.133	$\{E, C_{2z}\mathcal{T}, M_y\mathcal{T}, M_x, \mathcal{T}, C_{2z}, M_y, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	$p_{2a}mm2$ 23.7.135	$\{E, C_{2z}\mathcal{T}, M_y, M_x\mathcal{T}, \mathcal{T}, C_{2z}, M_y\mathcal{T}, M_x\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_{2a}m'm'2$ 23.5.133	$\{E, C_{2z}, M_y\mathcal{T}, M_x\mathcal{T}, \mathcal{T}, C_{2z}\mathcal{T}, M_y, M_x\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}112$ 3.4.11	$\{E, C_{2z}\mathcal{T}, \mathcal{T}, C_{2z}\}$	0	0
			$\mathrm{PL}_{(010)}$	$p_{2b}m'11$ 11.6.55	$\{E, M_y\mathcal{T}, \mathcal{T}, M_y\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_{2a}m11$ 11.4.53	$\{E, M_x\mathcal{T}, \mathcal{T}, M_x\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0
48 24.6.141 $p^{-1}_{2b}m^{-1}a^{1}2$ SI p. 40	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$ $\{-1\|E\}$ $\{-1\|C_{2z}\}$ $\{1\|M_y\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$p_{2b}m'a'2$ 24.6.141	$\{E, C_{2z}\mathcal{T}, M_y, M_x\mathcal{T}, \mathcal{T}, C_{2z}, M_y\mathcal{T}, M_x\}$	0	0
			$\mathrm{AX}_{[010]}$	$p_{2b}ma2$ 24.7.142	$\{E, C_{2z}\mathcal{T}, M_y\mathcal{T}, M_x, \mathcal{T}, C_{2z}, M_y, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_{2b}ma2$ 24.7.142	$\{E, C_{2z}, M_y, M_x, \mathcal{T}, C_{2z}\mathcal{T}, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}112$ 3.4.11	$\{E, C_{2z}\mathcal{T}, \mathcal{T}, C_{2z}\}$	0	0
			$\mathrm{PL}_{(010)}$	$p_{2a}b11$ 12.4.60	$\{E, M_y, \mathcal{T}, M_y\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_{2b}m11$ 11.5.54	$\{E, M_x, \mathcal{T}, M_x\mathcal{T}\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG (OG) symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
49 24.7.142 $p^1_{2b}m^1a^12$ SI p. 40	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|M_y\}$ $\{1\|M_x\}$ $\{-1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$p_{2b}ma2$ 24.7.142	$\{E, C_{2z}\mathcal{T}, M_y\mathcal{T}, M_x, \mathcal{T}, C_{2z}, M_y, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	$p_{2b}m'a'2$ 24.6.141	$\{E, C_{2z}\mathcal{T}, M_y, M_x\mathcal{T}, \mathcal{T}, C_{2z}, M_y\mathcal{T}, M_x\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_{2b}m'a'2$ 24.6.141	$\{E, C_{2z}, M_y\mathcal{T}, M_x\mathcal{T}, \mathcal{T}, C_{2z}\mathcal{T}, M_y, M_x\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}\bar{1}12$ 3.4.11	$\{E, C_{2z}\mathcal{T}, \mathcal{T}, C_{2z}\}$	0	0
			$\mathrm{PL}_{(010)}$	$p_{2a}b\bar{1}1$ 12.4.60	$\{E, M_y\mathcal{T}, \mathcal{T}, M_y\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_{2b}m'\bar{1}1$ 11.6.55	$\{E, M_x\mathcal{T}, \mathcal{T}, M_x\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0
50 26.5.151 $c^1_pm^1m^12$ SI p. 41	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|M_y\}$ $\{1\|M_x\}$ $\{-1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$c_pmm'2'$ 26.7.153	$\{E, C_{2z}\mathcal{T}, M_y\mathcal{T}, M_x, \mathcal{T}, C_{2z}, M_y, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	$c_pmm'2'$ 26.7.153	$\{E, C_{2z}\mathcal{T}, M_y, M_x\mathcal{T}, \mathcal{T}, C_{2z}, M_y\mathcal{T}, M_x\}$	0	0
			$\mathrm{AX}_{[001]}$	$c_pm'm'2$ 26.6.152	$\{E, C_{2z}, M_y\mathcal{T}, M_x\mathcal{T}, \mathcal{T}, C_{2z}\mathcal{T}, M_y, M_x\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}\bar{1}12$ 3.4.11	$\{E, C_{2z}\mathcal{T}, \mathcal{T}, C_{2z}\}$	0	0
			$\mathrm{PL}_{(010)}$	$c_pm'\bar{1}1$ 13.5.65	$\{E, M_y\mathcal{T}, \mathcal{T}, M_y\}$	0	0
			$\mathrm{PL}_{(100)}$	$c_pm'\bar{1}1$ 13.5.65	$\{E, M_x\mathcal{T}, \mathcal{T}, M_x\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0
51 26.6.152 $c_p^{-1}m^{-1}m^12$ SI p. 41	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$ $\{-1\|E\}$ $\{-1\|C_{2z}\}$ $\{1\|M_y\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$c_pmm'2'$ 26.7.153	$\{E, C_{2z}\mathcal{T}, M_y, M_x\mathcal{T}, \mathcal{T}, C_{2z}, M_y\mathcal{T}, M_x\}$	0	0
			$\mathrm{AX}_{[010]}$	$c_pmm'2'$ 26.7.153	$\{E, C_{2z}\mathcal{T}, M_y\mathcal{T}, M_x, \mathcal{T}, C_{2z}, M_y, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	c_pmm2 26.5.151	$\{E, C_{2z}, M_y, M_x, \mathcal{T}, C_{2z}\mathcal{T}, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}\bar{1}12$ 3.4.11	$\{E, C_{2z}\mathcal{T}, \mathcal{T}, C_{2z}\}$	0	0
			$\mathrm{PL}_{(010)}$	$c_pm\bar{1}1$ 13.4.64	$\{E, M_y, \mathcal{T}, M_y\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(100)}$	$c_pm\bar{1}1$ 13.4.64	$\{E, M_x, \mathcal{T}, M_x\mathcal{T}\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0
52 26.7.153 $c^1_pm^{-1}m^{-1}2$ SI p. 41	$T\tau$ AFM	$\{1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|M_y\}$ $\{1\|M_x\}$ $\{-1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|M_y\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	c_pmm2 26.5.151	$\{E, C_{2z}, M_y, M_x, \mathcal{T}, C_{2z}\mathcal{T}, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	$c_pm'm'2$ 26.6.152	$\{E, C_{2z}, M_y\mathcal{T}, M_x\mathcal{T}, \mathcal{T}, C_{2z}\mathcal{T}, M_y, M_x\}$	0	0
			$\mathrm{AX}_{[001]}$	$c_pmm'2'$ 26.7.153	$\{E, C_{2z}\mathcal{T}, M_y, M_x\mathcal{T}, \mathcal{T}, C_{2z}, M_y\mathcal{T}, M_x\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}\bar{1}12$ 3.4.11	$\{E, C_{2z}, \mathcal{T}, C_{2z}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	$c_pm\bar{1}1$ 13.4.64	$\{E, M_y, \mathcal{T}, M_y\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(100)}$	$c_pm'\bar{1}1$ 13.5.65	$\{E, M_x\mathcal{T}, \mathcal{T}, M_x\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG (OG) symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
53 27.6.159 $p^{-1}_{2b}m^{1}2^{-1}m$ SI p. 41	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2y}\}$ $\{-1\|M_x\}$ $\{-1\|M_z\}$ $\{-1\|E\}$ $\{-1\|C_{2y}\}$ $\{1\|M_x\}$ $\{1\|M_z\}$	$\mathrm{AX}_{[100]}$	$p_{2b}m'2'm$ 27.7.160	$\{E, C_{2y}\mathcal{T}, M_x\mathcal{T}, M_z, \mathcal{T}, C_{2y}, M_x, M_z\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	$p_{2b}m2m$ 27.13.166	$\{E, C_{2y}, M_x, M_z, \mathcal{T}, C_{2y}\mathcal{T}, M_x\mathcal{T}, M_z\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_{2b}m2'm'$ 27.8.161	$\{E, C_{2y}\mathcal{T}, M_x, M_z\mathcal{T}, \mathcal{T}, C_{2y}, M_x\mathcal{T}, M_z\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}11m$ 4.4.15	$\{E, M_z, \mathcal{T}, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	$p_{2a}2'11$ 8.4.37	$\{E, C_{2y}\mathcal{T}, \mathcal{T}, C_{2y}\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_{2b}m11$ 11.5.54	$\{E, M_x, \mathcal{T}, M_x\mathcal{T}\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0
54 27.7.160 $p^{-1}_{2b}m^{-1}2^{1}m$ SI p. 41	$T\tau$ AFM	$\{1\|E\}$ $\{-1\|C_{2y}\}$ $\{-1\|M_x\}$ $\{1\|M_z\}$ $\{-1\|E\}$ $\{1\|C_{2y}\}$ $\{1\|M_x\}$ $\{-1\|M_z\}$	$\mathrm{AX}_{[100]}$	$p_{2b}m'2m'$ 27.6.159	$\{E, C_{2y}, M_x\mathcal{T}, M_z\mathcal{T}, \mathcal{T}, C_{2y}\mathcal{T}, M_x, M_z\}$	0	0
			$\mathrm{AX}_{[010]}$	$p_{2b}m2'm'$ 27.8.161	$\{E, C_{2y}\mathcal{T}, M_x, M_z\mathcal{T}, \mathcal{T}, C_{2y}, M_x\mathcal{T}, M_z\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_{2b}m2m$ 27.13.166	$\{E, C_{2y}, M_x, M_z, \mathcal{T}, C_{2y}\mathcal{T}, M_x\mathcal{T}, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}11m'$ 4.5.16	$\{E, M_z\mathcal{T}, \mathcal{T}, M_z\}$	0	0
			$\mathrm{PL}_{(010)}$	$p_{2a}211$ 8.6.39	$\{E, C_{2y}, \mathcal{T}, C_{2y}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_{2b}m11$ 11.5.54	$\{E, M_x, \mathcal{T}, M_x\mathcal{T}\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0
55 27.8.161 $p^{1}_{2b}m^{-1}2^{-1}m$ SI p. 42	$T\tau$ AFM	$\{1\|E\}$ $\{-1\|C_{2y}\}$ $\{1\|M_x\}$ $\{-1\|M_z\}$ $\{-1\|E\}$ $\{1\|C_{2y}\}$ $\{-1\|M_x\}$ $\{1\|M_z\}$	$\mathrm{AX}_{[100]}$	$p_{2b}m2m$ 27.13.166	$\{E, C_{2y}, M_x, M_z, \mathcal{T}, C_{2y}\mathcal{T}, M_x\mathcal{T}, M_z\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	$p_{2b}m'2m$ 27.7.160	$\{E, C_{2y}\mathcal{T}, M_x\mathcal{T}, M_z, \mathcal{T}, C_{2y}, M_x, M_z\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_{2b}m'2m'$ 27.6.159	$\{E, C_{2y}, M_x\mathcal{T}, M_z\mathcal{T}, \mathcal{T}, C_{2y}\mathcal{T}, M_x, M_z\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}11m$ 4.4.15	$\{E, M_z, \mathcal{T}, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	$p_{2a}211$ 8.6.39	$\{E, C_{2y}, \mathcal{T}, C_{2y}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_{2b}m'11$ 11.6.55	$\{E, M_x\mathcal{T}, \mathcal{T}, M_x\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0
56 27.9.162 $p^{-1}_{2a}m^{1}2^{-1}m$ SI p. 42	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2y}\}$ $\{-1\|M_x\}$ $\{-1\|M_z\}$ $\{-1\|E\}$ $\{-1\|C_{2y}\}$ $\{1\|M_x\}$ $\{1\|M_z\}$	$\mathrm{AX}_{[100]}$	$p_{2a}m2m$ 27.12.165	$\{E, C_{2y}\mathcal{T}, M_x\mathcal{T}, M_z, \mathcal{T}, C_{2y}, M_x, M_z\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	$p_{2a}m2m$ 27.12.165	$\{E, C_{2y}, M_x, M_z, \mathcal{T}, C_{2y}\mathcal{T}, M_x\mathcal{T}, M_z\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_{2a}m'2m'$ 27.9.162	$\{E, C_{2y}\mathcal{T}, M_x, M_z\mathcal{T}, \mathcal{T}, C_{2y}, M_x\mathcal{T}, M_z\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}11m$ 4.4.15	$\{E, M_z, \mathcal{T}, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	$p_{2b}211$ 8.7.40	$\{E, C_{2y}\mathcal{T}, \mathcal{T}, C_{2y}\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_{2a}m11$ 11.4.53	$\{E, M_x, \mathcal{T}, M_x\mathcal{T}\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG (OG) symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
57 27.10.163 $p^1_c m^1 2^1 m$ SI p. 42	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2y}\}$ $\{1\|M_x\}$ $\{1\|M_z\}$ $\{-1\|E\}$ $\{-1\|C_{2y}\}$ $\{-1\|M_x\}$ $\{-1\|M_z\}$	$\mathrm{AX}_{[100]}$	$p_c m'2m'$ 27.11.164	$\{E, C_{2y}\mathcal{T}, M_x, M_z\mathcal{T}, \mathcal{T}, C_{2y}, M_x\mathcal{T}, M_z\}$	0	0
			$\mathrm{AX}_{[010]}$	$p_c m'2m'$ 27.11.164	$\{E, C_{2y}, M_x\mathcal{T}, M_z\mathcal{T}, \mathcal{T}, C_{2y}\mathcal{T}, M_x, M_z\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_c m2m$ 27.10.163	$\{E, C_{2y}\mathcal{T}, M_x\mathcal{T}, M_z, \mathcal{T}, C_{2y}, M_x, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}11m'$ 4.5.16	$\{E, M_z\mathcal{T}, \mathcal{T}, M_z\}$	0	0
			$\mathrm{PL}_{(010)}$	$p_c 211$ 8.5.38	$\{E, C_{2y}\mathcal{T}, \mathcal{T}, C_{2y}\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_c m11$ 11.7.56	$\{E, M_x\mathcal{T}, \mathcal{T}, M_x\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0
58 27.11.164 $p_c^{-1} m^1 2^{-1} m$ SI p. 42	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2y}\}$ $\{-1\|M_x\}$ $\{-1\|M_z\}$ $\{-1\|E\}$ $\{-1\|C_{2y}\}$ $\{1\|M_x\}$ $\{1\|M_z\}$	$\mathrm{AX}_{[100]}$	$p_c m2m$ 27.10.163	$\{E, C_{2y}\mathcal{T}, M_x\mathcal{T}, M_z, \mathcal{T}, C_{2y}, M_x, M_z\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	$p_c m2m$ 27.10.163	$\{E, C_{2y}, M_x, M_z, \mathcal{T}, C_{2y}\mathcal{T}, M_x\mathcal{T}, M_z\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_c m'2m'$ 27.11.164	$\{E, C_{2y}\mathcal{T}, M_x, M_z\mathcal{T}, \mathcal{T}, C_{2y}, M_x\mathcal{T}, M_z\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}11m$ 4.4.15	$\{E, M_z, \mathcal{T}, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	$p_c 211$ 8.5.38	$\{E, C_{2y}\mathcal{T}, \mathcal{T}, C_{2y}\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_c m11$ 11.7.56	$\{E, M_x, \mathcal{T}, M_x\mathcal{T}\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0
59 27.12.165 $p^1_{2a} m^1 2^1 m$ SI p. 42	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2y}\}$ $\{1\|M_x\}$ $\{1\|M_z\}$ $\{-1\|E\}$ $\{-1\|C_{2y}\}$ $\{-1\|M_x\}$ $\{-1\|M_z\}$	$\mathrm{AX}_{[100]}$	$p_{2a}m'2m'$ 27.9.162	$\{E, C_{2y}\mathcal{T}, M_x, M_z\mathcal{T}, \mathcal{T}, C_{2y}, M_x\mathcal{T}, M_z\}$	0	0
			$\mathrm{AX}_{[010]}$	$p_{2a}m'2m'$ 27.9.162	$\{E, C_{2y}, M_x\mathcal{T}, M_z\mathcal{T}, \mathcal{T}, C_{2y}\mathcal{T}, M_x, M_z\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_{2a}m2m$ 27.12.165	$\{E, C_{2y}\mathcal{T}, M_x\mathcal{T}, M_z, \mathcal{T}, C_{2y}, M_x, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}11m'$ 4.5.16	$\{E, M_z\mathcal{T}, \mathcal{T}, M_z\}$	0	0
			$\mathrm{PL}_{(010)}$	$p_{2b}211$ 8.7.40	$\{E, C_{2y}\mathcal{T}, \mathcal{T}, C_{2y}\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_{2a}m11$ 11.4.53	$\{E, M_x\mathcal{T}, \mathcal{T}, M_x\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0
60 27.13.166 $p^1_{2b} m^1 2^1 m$ SI p. 42	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2y}\}$ $\{1\|M_x\}$ $\{1\|M_z\}$ $\{-1\|E\}$ $\{-1\|C_{2y}\}$ $\{-1\|M_x\}$ $\{-1\|M_z\}$	$\mathrm{AX}_{[100]}$	$p_{2b}m2'm'$ 27.8.161	$\{E, C_{2y}\mathcal{T}, M_x, M_z\mathcal{T}, \mathcal{T}, C_{2y}, M_x\mathcal{T}, M_z\}$	0	0
			$\mathrm{AX}_{[010]}$	$p_{2b}m'2m'$ 27.6.159	$\{E, C_{2y}, M_x\mathcal{T}, M_z\mathcal{T}, \mathcal{T}, C_{2y}\mathcal{T}, M_x, M_z\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_{2b}m'2'm$ 27.7.160	$\{E, C_{2y}\mathcal{T}, M_x\mathcal{T}, M_z, \mathcal{T}, C_{2y}, M_x, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}11m'$ 4.5.16	$\{E, M_z\mathcal{T}, \mathcal{T}, M_z\}$	0	0
			$\mathrm{PL}_{(010)}$	$p_{2a}2'11$ 8.4.37	$\{E, C_{2y}\mathcal{T}, \mathcal{T}, C_{2y}\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_{2b}m'11$ 11.6.55	$\{E, M_x\mathcal{T}, \mathcal{T}, M_x\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG (OG) symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
61 28.6.172 $p^1_{2a}m^{-1}2_1^{-1}b$ SI p. 43	$T\tau$ AFM	$\{1\|E\}$ $\{-1\|C_{2y}\}$ $\{-1\|M_z\}$ $\{1\|M_x\}$ $\{-1\|E\}$ $\{1\|C_{2y}\}$ $\{1\|M_z\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$p_{2a}m2_1b$ 28.7.173	$\{E, C_{2y}, M_z, M_x, \mathcal{T}, C_{2y}\mathcal{T}, M_z\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	$p_{2a}m2_1b$ 28.7.173	$\{E, C_{2y}\mathcal{T}, M_z, M_x\mathcal{T}, \mathcal{T}, C_{2y}, M_z\mathcal{T}, M_x\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_{2a}m2_1'b'$ 28.6.172	$\{E, C_{2y}, M_z\mathcal{T}, M_x\mathcal{T}, \mathcal{T}, C_{2y}\mathcal{T}, M_z, M_x\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2b}11a$ 5.4.20	$\{E, M_z, \mathcal{T}, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	$p_{2b}2_111$ 9.4.44	$\{E, C_{2y}, \mathcal{T}, C_{2y}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_{2a}m11$ 11.4.53	$\{E, M_x\mathcal{T}, \mathcal{T}, M_x\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0
62 28.7.173 $p^1_{2a}m^12_1^1b$ SI p. 43	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2y}\}$ $\{1\|M_z\}$ $\{1\|M_x\}$ $\{-1\|E\}$ $\{-1\|C_{2y}\}$ $\{-1\|M_z\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$p_{2a}m2_1'b'$ 28.6.172	$\{E, C_{2y}\mathcal{T}, M_z\mathcal{T}, M_x, \mathcal{T}, C_{2y}, M_z, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	$p_{2a}m2_1'b'$ 28.6.172	$\{E, C_{2y}, M_z\mathcal{T}, M_x\mathcal{T}, \mathcal{T}, C_{2y}\mathcal{T}, M_z, M_x\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_{2a}m2_1b$ 28.7.173	$\{E, C_{2y}\mathcal{T}, M_z, M_x\mathcal{T}, \mathcal{T}, C_{2y}, M_z\mathcal{T}, M_x\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2b}11a$ 5.4.20	$\{E, M_z\mathcal{T}, \mathcal{T}, M_z\}$	0	0
			$\mathrm{PL}_{(010)}$	$p_{2b}2_111$ 9.4.44	$\{E, C_{2y}\mathcal{T}, \mathcal{T}, C_{2y}\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_{2a}m11$ 11.4.53	$\{E, M_x\mathcal{T}, \mathcal{T}, M_x\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0
63 29.6.179 $p^{-1}_{2a}b^12_1^{-1}m$ SI p. 43	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2y}\}$ $\{-1\|M_x\}$ $\{-1\|M_z\}$ $\{-1\|E\}$ $\{-1\|C_{2y}\}$ $\{1\|M_x\}$ $\{1\|M_z\}$	$\mathrm{AX}_{[100]}$	$p_{2a}b2_1m$ 29.7.180	$\{E, C_{2y}\mathcal{T}, M_x\mathcal{T}, M_z, \mathcal{T}, C_{2y}, M_x, M_z\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	$p_{2a}b2_1m$ 29.7.180	$\{E, C_{2y}, M_x, M_z, \mathcal{T}, C_{2y}\mathcal{T}, M_x\mathcal{T}, M_z\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_{2a}b'2_1m'$ 29.6.179	$\{E, C_{2y}\mathcal{T}, M_x, M_z\mathcal{T}, \mathcal{T}, C_{2y}, M_x\mathcal{T}, M_z\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}11m$ 4.4.15	$\{E, M_z, \mathcal{T}, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	$p_{2b}2_111$ 9.4.44	$\{E, C_{2y}\mathcal{T}, \mathcal{T}, C_{2y}\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_{2a}b11$ 12.4.60	$\{E, M_x, \mathcal{T}, M_x\mathcal{T}\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0
64 29.7.180 $p^1_{2a}b^12_1^1m$ SI p. 43	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2y}\}$ $\{1\|M_x\}$ $\{1\|M_z\}$ $\{-1\|E\}$ $\{-1\|C_{2y}\}$ $\{-1\|M_x\}$ $\{-1\|M_z\}$	$\mathrm{AX}_{[100]}$	$p_{2a}b'2_1m'$ 29.6.179	$\{E, C_{2y}\mathcal{T}, M_x, M_z\mathcal{T}, \mathcal{T}, C_{2y}, M_x\mathcal{T}, M_z\}$	0	0
			$\mathrm{AX}_{[010]}$	$p_{2a}b'2_1m'$ 29.6.179	$\{E, C_{2y}, M_x\mathcal{T}, M_z\mathcal{T}, \mathcal{T}, C_{2y}\mathcal{T}, M_x, M_z\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_{2a}b2_1m$ 29.7.180	$\{E, C_{2y}\mathcal{T}, M_x\mathcal{T}, M_z, \mathcal{T}, C_{2y}, M_x, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}11m'$ 4.5.16	$\{E, M_z\mathcal{T}, \mathcal{T}, M_z\}$	0	0
			$\mathrm{PL}_{(010)}$	$p_{2b}2_111$ 9.4.44	$\{E, C_{2y}\mathcal{T}, \mathcal{T}, C_{2y}\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_{2a}b11$ 12.4.60	$\{E, M_x\mathcal{T}, \mathcal{T}, M_x\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG (OG) symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
65 30.6.186 $p^{-1}_{2a}b^{1}2^{-1}b$ SI p. 43	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2y}\}$ $\{-1\|M_z\}$ $\{-1\|M_x\}$ $\{-1\|E\}$ $\{-1\|C_{2y}\}$ $\{1\|M_z\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$p_{2a}b2b$ 30.7.187	$\{E, C_{2y}\mathcal{T}, M_z, M_x\mathcal{T}, \mathcal{T}, C_{2y}, M_z\mathcal{T}, M_x\}$	0	0
			$\mathrm{AX}_{[010]}$	$p_{2a}b2b$ 30.7.187	$\{E, C_{2y}, M_z, M_x, \mathcal{T}, C_{2y}\mathcal{T}, M_z\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_{2a}b'2b'$ 30.6.186	$\{E, C_{2y}\mathcal{T}, M_z\mathcal{T}, M_x, \mathcal{T}, C_{2y}, M_z, M_x\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2b}11a$ 5.4.20	$\{E, M_z, \mathcal{T}, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	$p_{2b}211$ 8.7.40	$\{E, C_{2y}\mathcal{T}, \mathcal{T}, C_{2y}\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_{2a}b11$ 12.4.60	$\{E, M_x, \mathcal{T}, M_x\mathcal{T}\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0
66 30.7.187 $p^{1}_{2a}b^{1}2^{1}b$ SI p. 44	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2y}\}$ $\{1\|M_z\}$ $\{1\|M_x\}$ $\{-1\|E\}$ $\{-1\|C_{2y}\}$ $\{-1\|M_z\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$p_{2a}b'2b'$ 30.6.186	$\{E, C_{2y}\mathcal{T}, M_z\mathcal{T}, M_x, \mathcal{T}, C_{2y}, M_z, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	$p_{2a}b'2b'$ 30.6.186	$\{E, C_{2y}, M_z\mathcal{T}, M_x\mathcal{T}, \mathcal{T}, C_{2y}\mathcal{T}, M_z, M_x\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_{2a}b2b$ 30.7.187	$\{E, C_{2y}\mathcal{T}, M_z, M_x\mathcal{T}, \mathcal{T}, C_{2y}, M_z\mathcal{T}, M_x\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2b}11a$ 5.4.20	$\{E, M_z\mathcal{T}, \mathcal{T}, M_z\}$	0	0
			$\mathrm{PL}_{(010)}$	$p_{2b}211$ 8.7.40	$\{E, C_{2y}\mathcal{T}, \mathcal{T}, C_{2y}\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_{2a}b11$ 12.4.60	$\{E, M_x\mathcal{T}, \mathcal{T}, M_x\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0
67 31.6.193 $p^{1}_{2b}m^{-1}2^{-1}a$ SI p. 44	$T\tau$ AFM	$\{1\|E\}$ $\{-1\|C_{2y}\}$ $\{-1\|M_z\}$ $\{1\|M_x\}$ $\{-1\|E\}$ $\{1\|C_{2y}\}$ $\{1\|M_z\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$p_{2b}m2a$ 31.9.196	$\{E, C_{2y}, M_z, M_x, \mathcal{T}, C_{2y}\mathcal{T}, M_z\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	$p_{2b}m'2'a$ 31.7.194	$\{E, C_{2y}\mathcal{T}, M_z, M_x\mathcal{T}, \mathcal{T}, C_{2y}, M_z\mathcal{T}, M_x\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_{2b}m'2a'$ 31.8.195	$\{E, C_{2y}, M_z\mathcal{T}, M_x\mathcal{T}, \mathcal{T}, C_{2y}\mathcal{T}, M_z, M_x\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2b}11a$ 5.4.20	$\{E, M_z, \mathcal{T}, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	$p_{2a}211$ 8.6.39	$\{E, C_{2y}, \mathcal{T}, C_{2y}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_{2b}m'11$ 11.6.55	$\{E, M_x\mathcal{T}, \mathcal{T}, M_x\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0
68 31.7.194 $p^{-1}_{2b}m^{-1}2^{1}a$ SI p. 44	$T\tau$ AFM	$\{1\|E\}$ $\{-1\|C_{2y}\}$ $\{1\|M_z\}$ $\{-1\|M_x\}$ $\{-1\|E\}$ $\{1\|C_{2y}\}$ $\{-1\|M_z\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$p_{2b}m'2a'$ 31.8.195	$\{E, C_{2y}, M_z\mathcal{T}, M_x\mathcal{T}, \mathcal{T}, C_{2y}\mathcal{T}, M_z, M_x\}$	0	0
			$\mathrm{AX}_{[010]}$	$p_{2b}m2'a'$ 31.6.193	$\{E, C_{2y}\mathcal{T}, M_z\mathcal{T}, M_x, \mathcal{T}, C_{2y}, M_z, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_{2b}m2a$ 31.9.196	$\{E, C_{2y}, M_z, M_x, \mathcal{T}, C_{2y}\mathcal{T}, M_z\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2b}11a$ 5.4.20	$\{E, M_z\mathcal{T}, \mathcal{T}, M_z\}$	0	0
			$\mathrm{PL}_{(010)}$	$p_{2a}211$ 8.6.39	$\{E, C_{2y}, \mathcal{T}, C_{2y}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_{2b}m11$ 11.5.54	$\{E, M_x, \mathcal{T}, M_x\mathcal{T}\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG (OG) symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
69 31.8.195 $p^{-1}_{2b}m^{1}2^{-1}a$ SI p. 44	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2y}\}$ $\{-1\|M_z\}$ $\{-1\|M_x\}$ $\{-1\|E\}$ $\{-1\|C_{2y}\}$ $\{1\|M_z\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$p_{2b}m'2'a$ 31.7.194	$\{E, C_{2y}\mathcal{T}, M_z, M_x\mathcal{T}, \mathcal{T}, C_{2y}, M_z\mathcal{T}, M_x\}$	0	0
			$\mathrm{AX}_{[010]}$	$p_{2b}m2a$ 31.9.196	$\{E, C_{2y}, M_z, M_x, \mathcal{T}, C_{2y}\mathcal{T}, M_z\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_{2b}m2'a'$ 31.6.193	$\{E, C_{2y}\mathcal{T}, M_z\mathcal{T}, M_x, \mathcal{T}, C_{2y}, M_z, M_x\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2b}11a$ 5.4.20	$\{E, M_z, \mathcal{T}, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	$p_{2a}2'11$ 8.4.37	$\{E, C_{2y}\mathcal{T}, \mathcal{T}, C_{2y}\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_{2b}m11$ 11.5.54	$\{E, M_x, \mathcal{T}, M_x\mathcal{T}\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0
70 31.9.196 $p^{1}_{2b}m^{1}2^{1}a$ SI p. 44	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2y}\}$ $\{1\|M_z\}$ $\{1\|M_x\}$ $\{-1\|E\}$ $\{-1\|C_{2y}\}$ $\{-1\|M_z\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$p_{2b}m2'a'$ 31.6.193	$\{E, C_{2y}\mathcal{T}, M_z\mathcal{T}, M_x, \mathcal{T}, C_{2y}, M_z, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	$p_{2b}m'2a'$ 31.8.195	$\{E, C_{2y}, M_z\mathcal{T}, M_x\mathcal{T}, \mathcal{T}, C_{2y}\mathcal{T}, M_z, M_x\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_{2b}m'2'a$ 31.7.194	$\{E, C_{2y}\mathcal{T}, M_z, M_x\mathcal{T}, \mathcal{T}, C_{2y}, M_z\mathcal{T}, M_x\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2b}11a$ 5.4.20	$\{E, M_z\mathcal{T}, \mathcal{T}, M_z\}$	0	0
			$\mathrm{PL}_{(010)}$	$p_{2a}2'11$ 8.4.37	$\{E, C_{2y}\mathcal{T}, \mathcal{T}, C_{2y}\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_{2b}m'11$ 11.6.55	$\{E, M_x\mathcal{T}, \mathcal{T}, M_x\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0
71 35.6.217 $c^{1}_{p}m^{1}2^{1}m$ SI p. 44	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2y}\}$ $\{1\|M_x\}$ $\{1\|M_z\}$ $\{-1\|E\}$ $\{-1\|C_{2y}\}$ $\{-1\|M_x\}$ $\{-1\|M_z\}$	$\mathrm{AX}_{[100]}$	$c_pm2'm'$ 35.8.219	$\{E, C_{2y}\mathcal{T}, M_x, M_z\mathcal{T}, \mathcal{T}, C_{2y}, M_x\mathcal{T}, M_z\}$	0	0
			$\mathrm{AX}_{[010]}$	$c_pm'2m'$ 35.7.218	$\{E, C_{2y}, M_x\mathcal{T}, M_z\mathcal{T}, \mathcal{T}, C_{2y}\mathcal{T}, M_x, M_z\}$	0	0
			$\mathrm{AX}_{[001]}$	$c_pm'2'm$ 35.9.220	$\{E, C_{2y}\mathcal{T}, M_x\mathcal{T}, M_z, \mathcal{T}, C_{2y}, M_x, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}11m'$ 4.5.16	$\{E, M_z\mathcal{T}, \mathcal{T}, M_z\}$	0	0
			$\mathrm{PL}_{(010)}$	$c_p2'11$ 10.5.49	$\{E, C_{2y}\mathcal{T}, \mathcal{T}, C_{2y}\}$	0	0
			$\mathrm{PL}_{(100)}$	$c_pm'11$ 13.5.65	$\{E, M_x\mathcal{T}, \mathcal{T}, M_x\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0
72 35.7.218 $c^{-1}_{p}m^{1}2^{-1}m$ SI p. 45	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2y}\}$ $\{-1\|M_x\}$ $\{-1\|M_z\}$ $\{-1\|E\}$ $\{-1\|C_{2y}\}$ $\{1\|M_x\}$ $\{1\|M_z\}$	$\mathrm{AX}_{[100]}$	$c_pm'2'm$ 35.9.220	$\{E, C_{2y}\mathcal{T}, M_x\mathcal{T}, M_z, \mathcal{T}, C_{2y}, M_x, M_z\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	c_pm2m 35.6.217	$\{E, C_{2y}, M_x, M_z, \mathcal{T}, C_{2y}\mathcal{T}, M_x\mathcal{T}, M_z\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$c_pm2'm'$ 35.8.219	$\{E, C_{2y}\mathcal{T}, M_x, M_z\mathcal{T}, \mathcal{T}, C_{2y}, M_x\mathcal{T}, M_z\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}11m$ 4.4.15	$\{E, M_z, \mathcal{T}, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	$c_p2'11$ 10.5.49	$\{E, C_{2y}\mathcal{T}, \mathcal{T}, C_{2y}\}$	0	0
			$\mathrm{PL}_{(100)}$	c_pm11 13.4.64	$\{E, M_x, \mathcal{T}, M_x\mathcal{T}\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG (OG) symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
73 35.8.219 $c_p^1m^{-1}2^{-1}m$ SI p. 45	$T\tau$ AFM	$\{1\|E\}$ $\{-1\|C_{2y}\}$ $\{1\|M_x\}$ $\{-1\|M_z\}$ $\{-1\|E\}$ $\{1\|C_{2y}\}$ $\{-1\|M_x\}$ $\{1\|M_z\}$	$\mathrm{AX}_{[100]}$	c_pm2m 35.6.217	$\{E, C_{2y}, M_x, M_z, \mathcal{T}, C_{2y}\mathcal{T}, M_x\mathcal{T}, M_z\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	$c_pm'2'm$ 35.9.220	$\{E, C_{2y}\mathcal{T}, M_x\mathcal{T}, M_z, \mathcal{T}, C_{2y}, M_x, M_z\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$c_pm'2m'$ 35.7.218	$\{E, C_{2y}, M_x\mathcal{T}, M_z\mathcal{T}, \mathcal{T}, C_{2y}\mathcal{T}, M_x, M_z\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}11m$ 4.4.15	$\{E, M_z, \mathcal{T}, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	c_p211 10.4.48	$\{E, C_{2y}, \mathcal{T}, C_{2y}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(100)}$	$c_pm'11$ 13.5.65	$\{E, M_x\mathcal{T}, \mathcal{T}, M_x\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0
74 35.9.220 $c_p^{-1}m^{-1}2^1m$ SI p. 45	$T\tau$ AFM	$\{1\|E\}$ $\{-1\|C_{2y}\}$ $\{-1\|M_x\}$ $\{1\|M_z\}$ $\{-1\|E\}$ $\{1\|C_{2y}\}$ $\{1\|M_x\}$ $\{-1\|M_z\}$	$\mathrm{AX}_{[100]}$	$c_pm'2m'$ 35.7.218	$\{E, C_{2y}, M_x\mathcal{T}, M_z\mathcal{T}, \mathcal{T}, C_{2y}\mathcal{T}, M_x, M_z\}$	0	0
			$\mathrm{AX}_{[010]}$	$c_pm2'm'$ 35.8.219	$\{E, C_{2y}\mathcal{T}, M_x, M_z\mathcal{T}, \mathcal{T}, C_{2y}, M_x\mathcal{T}, M_z\}$	0	0
			$\mathrm{AX}_{[001]}$	c_pm2m 35.6.217	$\{E, C_{2y}, M_x, M_z, \mathcal{T}, C_{2y}\mathcal{T}, M_x\mathcal{T}, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}11m'$ 4.5.16	$\{E, M_z\mathcal{T}, \mathcal{T}, M_z\}$	0	0
			$\mathrm{PL}_{(010)}$	c_p211 10.4.48	$\{E, C_{2y}, \mathcal{T}, C_{2y}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(100)}$	c_pm11 13.4.64	$\{E, M_x, \mathcal{T}, M_x\mathcal{T}\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0
75 36.6.226 $c_p^1m^12^1a$ SI p. 45	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2y}\}$ $\{1\|M_x\}$ $\{1\|M_z\}$ $\{-1\|E\}$ $\{-1\|C_{2y}\}$ $\{-1\|M_x\}$ $\{-1\|M_z\}$	$\mathrm{AX}_{[100]}$	$c_pm2'a'$ 36.8.228	$\{E, C_{2y}\mathcal{T}, M_x, M_z\mathcal{T}, \mathcal{T}, C_{2y}, M_x\mathcal{T}, M_z\}$	0	0
			$\mathrm{AX}_{[010]}$	$c_pm'2a'$ 36.7.227	$\{E, C_{2y}, M_x\mathcal{T}, M_z\mathcal{T}, \mathcal{T}, C_{2y}\mathcal{T}, M_x, M_z\}$	0	0
			$\mathrm{AX}_{[001]}$	$c_pm'2'a$ 36.9.229	$\{E, C_{2y}\mathcal{T}, M_x\mathcal{T}, M_z, \mathcal{T}, C_{2y}, M_x, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2b}11a$ 5.4.20	$\{E, M_z\mathcal{T}, \mathcal{T}, M_z\}$	0	0
			$\mathrm{PL}_{(010)}$	$c_p2'11$ 10.5.49	$\{E, C_{2y}\mathcal{T}, \mathcal{T}, C_{2y}\}$	0	0
			$\mathrm{PL}_{(100)}$	$c_pm'11$ 13.5.65	$\{E, M_x\mathcal{T}, \mathcal{T}, M_x\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0
76 36.7.227 $c_p^{-1}m^12^{-1}a$ SI p. 45	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2y}\}$ $\{-1\|M_x\}$ $\{-1\|M_z\}$ $\{-1\|E\}$ $\{-1\|C_{2y}\}$ $\{1\|M_x\}$ $\{1\|M_z\}$	$\mathrm{AX}_{[100]}$	$c_pm'2'a$ 36.9.229	$\{E, C_{2y}\mathcal{T}, M_x\mathcal{T}, M_z, \mathcal{T}, C_{2y}, M_x, M_z\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	c_pm2a 36.6.226	$\{E, C_{2y}, M_x, M_z, \mathcal{T}, C_{2y}\mathcal{T}, M_x\mathcal{T}, M_z\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$c_pm2'a'$ 36.8.228	$\{E, C_{2y}\mathcal{T}, M_x, M_z\mathcal{T}, \mathcal{T}, C_{2y}, M_x\mathcal{T}, M_z\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2b}11a$ 5.4.20	$\{E, M_z, \mathcal{T}, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	$c_p2'11$ 10.5.49	$\{E, C_{2y}\mathcal{T}, \mathcal{T}, C_{2y}\}$	0	0
			$\mathrm{PL}_{(100)}$	c_pm11 13.4.64	$\{E, M_x, \mathcal{T}, M_x\mathcal{T}\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG (OG) symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
77 36.8.228 $c_p^1m^{-1}2^{-1}a$ SI p. 46	$T\tau$ AFM	$\{1\|E\}$ $\{-1\|C_{2y}\}$ $\{1\|M_x\}$ $\{-1\|M_z\}$ $\{-1\|E\}$ $\{1\|C_{2y}\}$ $\{-1\|M_x\}$ $\{1\|M_z\}$	$\mathrm{AX}_{[100]}$	c_pm2a 36.6.226	$\{E,\ C_{2y},\ M_x,\ M_z,\ \mathcal{T},\ C_{2y}\mathcal{T},\ M_x\mathcal{T},\ M_z\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	$c_pm'2'a$ 36.9.229	$\{E,\ C_{2y}\mathcal{T},\ M_x\mathcal{T},\ M_z,\ \mathcal{T},\ C_{2y},\ M_x,\ M_z\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$c_pm'2a'$ 36.7.227	$\{E,\ C_{2y},\ M_x\mathcal{T},\ M_z\mathcal{T},\ \mathcal{T},\ C_{2y}\mathcal{T},\ M_x,\ M_z\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2b}11a$ 5.4.20	$\{E,\ M_z,\ \mathcal{T},\ M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	c_p211 10.4.48	$\{E,\ C_{2y},\ \mathcal{T},\ C_{2y}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(100)}$	$c_pm'11$ 13.5.65	$\{E,\ M_x\mathcal{T},\ \mathcal{T},\ M_x\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E,\ \mathcal{T}\}$	0	0
78 36.9.229 $c_p^{-1}m^{-1}2^1a$ SI p. 46	$T\tau$ AFM	$\{1\|E\}$ $\{-1\|C_{2y}\}$ $\{-1\|M_x\}$ $\{1\|M_z\}$ $\{-1\|E\}$ $\{1\|C_{2y}\}$ $\{1\|M_x\}$ $\{-1\|M_z\}$	$\mathrm{AX}_{[100]}$	$c_pm'2a'$ 36.7.227	$\{E,\ C_{2y},\ M_x\mathcal{T},\ M_z\mathcal{T},\ \mathcal{T},\ C_{2y}\mathcal{T},\ M_x,\ M_z\}$	0	0
			$\mathrm{AX}_{[010]}$	$c_pm2'a'$ 36.8.228	$\{E,\ C_{2y}\mathcal{T},\ M_x,\ M_z\mathcal{T},\ \mathcal{T},\ C_{2y},\ M_x\mathcal{T},\ M_z\}$	0	0
			$\mathrm{AX}_{[001]}$	c_pm2a 36.6.226	$\{E,\ C_{2y},\ M_x,\ M_z,\ \mathcal{T},\ C_{2y}\mathcal{T},\ M_x\mathcal{T},\ M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2b}11a$ 5.4.20	$\{E,\ M_z\mathcal{T},\ \mathcal{T},\ M_z\}$	0	0
			$\mathrm{PL}_{(010)}$	c_p211 10.4.48	$\{E,\ C_{2y},\ \mathcal{T},\ C_{2y}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(100)}$	c_pm11 13.4.64	$\{E,\ M_x,\ \mathcal{T},\ M_x\mathcal{T}\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E,\ \mathcal{T}\}$	0	0
79 37.8.237 $p_{2a}^1m^{-1}m^{-1}m$ SI p. 46	$T\tau$ AFM	$\{1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{-1\|M_y\}$ $\{1\|M_x\}$ $\{-1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{1\|M_y\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$p_{2a}mmm$ 37.13.242	$\{E,\ C_{2z},\ C_{2y},\ C_{2x},\ \mathcal{P},\ M_z,\ M_y,\ M_x,\ \mathcal{T},\ C_{2z}\mathcal{T},\ C_{2y}\mathcal{T},\ C_{2x}\mathcal{T},\ \mathcal{P}\mathcal{T},\ M_z\mathcal{T},\ M_y\mathcal{T},\ M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	$p_{2a}m'm'm$ 37.9.238	$\{E,\ C_{2z},\ C_{2y}\mathcal{T},\ C_{2x}\mathcal{T},\ \mathcal{P},\ M_z,\ M_y\mathcal{T},\ M_x\mathcal{T},\ \mathcal{T},\ C_{2z}\mathcal{T},\ C_{2y},\ C_{2x},\ \mathcal{P}\mathcal{T},\ M_z\mathcal{T},\ M_y,\ M_x\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_{2a}m'mm'$ 37.10.239	$\{E,\ C_{2z}\mathcal{T},\ C_{2y},\ C_{2x}\mathcal{T},\ \mathcal{P},\ M_z\mathcal{T},\ M_y,\ M_x\mathcal{T},\ \mathcal{T},\ C_{2z},\ C_{2y}\mathcal{T},\ C_{2x},\ \mathcal{P}\mathcal{T},\ M_z,\ M_y\mathcal{T},\ M_x\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}112/m$ 6.6.26	$\{E,\ C_{2z},\ \mathcal{P},\ M_z,\ \mathcal{T},\ C_{2z}\mathcal{T},\ \mathcal{P}\mathcal{T},\ M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	$p_{2b}2/m11$ 14.10.75	$\{E,\ C_{2y},\ \mathcal{P},\ M_y,\ \mathcal{T},\ C_{2y}\mathcal{T},\ \mathcal{P}\mathcal{T},\ M_y\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_{2a}2'/m'11$ 14.6.71	$\{E,\ C_{2x}\mathcal{T},\ \mathcal{P},\ M_x\mathcal{T},\ \mathcal{T},\ C_{2x},\ \mathcal{P}\mathcal{T},\ M_x\}$	0	0
			GEN	$p_{2a}\bar{1}$ 2.4.7	$\{E,\ \mathcal{P},\ \mathcal{T},\ \mathcal{P}\mathcal{T}\}$	0	0

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG (OG) symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
80 37.9.238 $p^{-1}_{2a}m^{-1}m^{1}m$ SI p. 46	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$ $\{-1\|E\}$ $\{-1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{1\|M_y\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$p_{2a}m'mm'$ 37.10.239	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}, \mathcal{T}, C_{2z}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_z, M_y\mathcal{T}, M_x\}$	0	0
			$\mathrm{AX}_{[010]}$	$p_{2a}mm'm'$ 37.8.237	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}, M_z\mathcal{T}, M_y\mathcal{T}, M_x, \mathcal{T}, C_{2z}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z, M_y, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_{2a}mmm$ 37.13.242	$\{E, C_{2z}, C_{2y}, C_{2x}, \mathcal{P}, M_z, M_y, M_x, \mathcal{T}, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}112'/m'$ 6.7.27	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, \mathcal{T}, C_{2z}, \mathcal{P}\mathcal{T}, M_z\}$	0	0
			$\mathrm{PL}_{(010)}$	$p_{2b}2/m11$ 14.10.75	$\{E, C_{2y}, \mathcal{P}, M_y, \mathcal{T}, C_{2y}\mathcal{T}, \mathcal{P}\mathcal{T}, M_y\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_{2a}2/m11$ 14.9.74	$\{E, C_{2x}, \mathcal{P}, M_x, \mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_x\mathcal{T}\}$	0	0
			GEN	$p_{2a}\bar{1}$ 2.4.7	$\{E, \mathcal{P}, \mathcal{T}, \mathcal{P}\mathcal{T}\}$	0	0
81 37.10.239 $p^{-1}_{2a}m^{1}m^{-1}m$ SI p. 47	$T\tau$ AFM	$\{1\|E\}$ $\{-1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{1\|M_y\}$ $\{-1\|M_x\}$ $\{-1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{-1\|M_y\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$p_{2a}m'm'm$ 37.9.238	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}, M_z, M_y\mathcal{T}, M_x\mathcal{T}, \mathcal{T}, C_{2z}\mathcal{T}, C_{2y}, C_{2x}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, M_y, M_x\}$	0	0
			$\mathrm{AX}_{[010]}$	$p_{2a}mmm$ 37.13.242	$\{E, C_{2z}, C_{2y}, C_{2x}, \mathcal{P}, M_z, M_y, M_x, \mathcal{T}, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_{2a}mm'm'$ 37.8.237	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}, M_z\mathcal{T}, M_y\mathcal{T}, M_x, \mathcal{T}, C_{2z}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z, M_y, M_x\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}112/m$ 6.6.26	$\{E, C_{2z}, \mathcal{P}, M_z, \mathcal{T}, C_{2z}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	$p_{2b}2'/m'11$ 14.7.72	$\{E, C_{2y}\mathcal{T}, \mathcal{P}, M_y\mathcal{T}, \mathcal{T}, C_{2y}, \mathcal{P}\mathcal{T}, M_y\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_{2a}2/m11$ 14.9.74	$\{E, C_{2x}, \mathcal{P}, M_x, \mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_x\mathcal{T}\}$	0	0
			GEN	$p_{2a}\bar{1}$ 2.4.7	$\{E, \mathcal{P}, \mathcal{T}, \mathcal{P}\mathcal{T}\}$	0	0

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG (OG) symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
82 37.11.240 $p^1_c m^1 m^1 m$ SI p. 47	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{1\|M_y\}$ $\{1\|M_x\}$ $\{-1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$p_c mm'm'$ 37.12.241	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}, M_z\mathcal{T}, M_y\mathcal{T}, M_x, \mathcal{T}, C_{2z}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z, M_y, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	$p_c mm'm'$ 37.12.241	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}, \mathcal{T}, C_{2z}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_z, M_y\mathcal{T}, M_x\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_c mmm$ 37.11.240	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}, M_z, M_y\mathcal{T}, M_x\mathcal{T}, \mathcal{T}, C_{2z}\mathcal{T}, C_{2y}, C_{2x}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, M_y, M_x\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}112'/m'$ 6.7.27	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, \mathcal{T}, C_{2z}, \mathcal{P}\mathcal{T}, M_z\}$	0	0
			$\mathrm{PL}_{(010)}$	$p_c2/m11$ 14.8.73	$\{E, C_{2y}\mathcal{T}, \mathcal{P}, M_y\mathcal{T}, \mathcal{T}, C_{2y}, \mathcal{P}\mathcal{T}, M_y\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_c2/m11$ 14.8.73	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}, \mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_x\}$	0	0
			GEN	$p_{2a}\bar{1}$ 2.4.7	$\{E, \mathcal{P}, \mathcal{T}, \mathcal{P}\mathcal{T}\}$	0	0
83 37.12.241 $p^1_c m^{-1} m^{-1} m$ SI p. 47	$T\tau$ AFM	$\{1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{-1\|M_y\}$ $\{1\|M_x\}$ $\{-1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{1\|M_y\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$p_c mmm$ 37.11.240	$\{E, C_{2z}, C_{2y}, C_{2x}, \mathcal{P}, M_z, M_y, M_x, \mathcal{T}, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	$p_c mmm$ 37.11.240	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}, M_z, M_y\mathcal{T}, M_x\mathcal{T}, \mathcal{T}, C_{2z}\mathcal{T}, C_{2y}, C_{2x}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, M_y, M_x\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_c mm'm'$ 37.12.241	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}, \mathcal{T}, C_{2z}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_z, M_y\mathcal{T}, M_x\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}112/m$ 6.6.26	$\{E, C_{2z}, \mathcal{P}, M_z, \mathcal{T}, C_{2z}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	$p_c2/m11$ 14.8.73	$\{E, C_{2y}, \mathcal{P}, M_y, \mathcal{T}, C_{2y}\mathcal{T}, \mathcal{P}\mathcal{T}, M_y\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_c2/m11$ 14.8.73	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}, \mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_x\}$	0	0
			GEN	$p_{2a}\bar{1}$ 2.4.7	$\{E, \mathcal{P}, \mathcal{T}, \mathcal{P}\mathcal{T}\}$	0	0

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG (OG) symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
84 37.13.242 $p^1_{2a}m^1m^1m$ SI p. 48	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{1\|M_y\}$ $\{1\|M_x\}$ $\{-1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$p_{2a}mm'm'$ 37.8.237	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}, M_z\mathcal{T}, M_y\mathcal{T}, M_x, \mathcal{T}, C_{2z}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z, M_y, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	$p_{2a}m'mm'$ 37.10.239	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}, \mathcal{T}, C_{2z}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_z, M_y\mathcal{T}, M_x\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_{2a}m'm'm$ 37.9.238	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}, M_z, M_y\mathcal{T}, M_x\mathcal{T}, \mathcal{T}, C_{2z}\mathcal{T}, C_{2y}, C_{2x}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, M_y, M_x\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}112'/m'$ 6.7.27	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, \mathcal{T}, C_{2z}, \mathcal{P}\mathcal{T}, M_z\}$	0	0
			$\mathrm{PL}_{(010)}$	$p_{2b}2'/m'11$ 14.7.72	$\{E, C_{2y}\mathcal{T}, \mathcal{P}, M_y\mathcal{T}, \mathcal{T}, C_{2y}, \mathcal{P}\mathcal{T}, M_y\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_{2a}2'/m'11$ 14.6.71	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}, \mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_x\}$	0	0
			GEN	$p_{2a}\bar{1}$ 2.4.7	$\{E, \mathcal{P}, \mathcal{T}, \mathcal{P}\mathcal{T}\}$	0	0
85 38.10.252 $p^{-1}_{2b}m^1a^{-1}a$ SI p. 48	$T\tau$ AFM	$\{1\|E\}$ $\{-1\|C_{2x}\}$ $\{-1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{1\|\mathcal{P}\}$ $\{-1\|M_x\}$ $\{-1\|M_z\}$ $\{1\|M_y\}$ $\{-1\|E\}$ $\{1\|C_{2x}\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{-1\|\mathcal{P}\}$ $\{1\|M_x\}$ $\{1\|M_z\}$ $\{-1\|M_y\}$	$\mathrm{AX}_{[100]}$	$p_{2b}m'a'a$ 38.12.254	$\{E, C_{2x}\mathcal{T}, C_{2z}, C_{2y}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}, M_z, M_y\mathcal{T}, \mathcal{T}, C_{2x}, C_{2z}\mathcal{T}, C_{2y}, \mathcal{P}\mathcal{T}, M_x, M_z\mathcal{T}, M_y\}$	0	0
			$\mathrm{AX}_{[010]}$	$p_{2b}maa$ 38.13.255	$\{E, C_{2x}, C_{2z}, C_{2y}, \mathcal{P}, M_x, M_z, M_y, \mathcal{T}, C_{2x}\mathcal{T}, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, \mathcal{P}\mathcal{T}, M_x\mathcal{T}, M_z\mathcal{T}, M_y\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_{2b}ma'a'$ 38.11.253	$\{E, C_{2x}, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, \mathcal{P}, M_x, M_z\mathcal{T}, M_y\mathcal{T}, \mathcal{T}, C_{2x}\mathcal{T}, C_{2z}, C_{2y}, \mathcal{P}\mathcal{T}, M_x\mathcal{T}, M_z, M_y\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2b}112/a$ 7.6.33	$\{E, C_{2z}, \mathcal{P}, M_z, \mathcal{T}, C_{2z}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	$p_{2a}2'/b'11$ 16.6.88	$\{E, C_{2y}\mathcal{T}, \mathcal{P}, M_y\mathcal{T}, \mathcal{T}, C_{2y}, \mathcal{P}\mathcal{T}, M_y\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_{2b}2/m11$ 14.10.75	$\{E, C_{2x}, \mathcal{P}, M_x, \mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_x\mathcal{T}\}$	0	0
			GEN	$p_{2a}\bar{1}$ 2.4.7	$\{E, \mathcal{P}, \mathcal{T}, \mathcal{P}\mathcal{T}\}$	0	0

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG (OG) symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
86 38.11.253 $p^{1}_{2b}m^{-1}a^{-1}a$ SI p. 48	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2x}\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_x\}$ $\{-1\|M_z\}$ $\{-1\|M_y\}$ $\{-1\|E\}$ $\{-1\|C_{2x}\}$ $\{1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_x\}$ $\{1\|M_z\}$ $\{1\|M_y\}$	$\mathrm{AX}_{[100]}$	$p_{2b}maa$ 38.13.255	$\{E, C_{2x}, C_{2z}, C_{2y}, \mathcal{P}, M_x, M_z, M_y, \mathcal{T}, C_{2x}\mathcal{T}, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, \mathcal{PT}, M_x\mathcal{T}, M_z\mathcal{T}, M_y\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	$p_{2b}m'a'a$ 38.12.254	$\{E, C_{2x}\mathcal{T}, C_{2z}, C_{2y}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}, M_z, M_y\mathcal{T}, \mathcal{T}, C_{2x}, C_{2z}\mathcal{T}, C_{2y}, \mathcal{PT}, M_x, M_z\mathcal{T}, M_y\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_{2b}m'aa'$ 38.10.252	$\{E, C_{2x}\mathcal{T}, C_{2z}\mathcal{T}, C_{2y}, \mathcal{P}, M_x\mathcal{T}, M_z\mathcal{T}, M_y, \mathcal{T}, C_{2x}, C_{2z}, C_{2y}\mathcal{T}, \mathcal{PT}, M_x, M_z, M_y\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2b}112/a$ 7.6.33	$\{E, C_{2z}, \mathcal{P}, M_z, \mathcal{T}, C_{2z}\mathcal{T}, \mathcal{PT}, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	$p_{2a}2/b11$ 16.7.89	$\{E, C_{2y}, \mathcal{P}, M_y, \mathcal{T}, C_{2y}\mathcal{T}, \mathcal{PT}, M_y\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_{2b}2'/m'11$ 14.7.72	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}, \mathcal{T}, C_{2x}, \mathcal{PT}, M_x\}$	0	0
			GEN	$p_{2a}\bar{1}$ 2.4.7	$\{E, \mathcal{P}, \mathcal{T}, \mathcal{PT}\}$	0	0
87 38.12.254 $p^{-1}_{2b}m^{-1}a^{1}a$ SI p. 49	$T\tau$ AFM	$\{1\|E\}$ $\{-1\|C_{2x}\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{1\|\mathcal{P}\}$ $\{-1\|M_x\}$ $\{1\|M_z\}$ $\{-1\|M_y\}$ $\{-1\|E\}$ $\{1\|C_{2x}\}$ $\{-1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{-1\|\mathcal{P}\}$ $\{1\|M_x\}$ $\{-1\|M_z\}$ $\{1\|M_y\}$	$\mathrm{AX}_{[100]}$	$p_{2b}m'aa'$ 38.10.252	$\{E, C_{2x}\mathcal{T}, C_{2z}\mathcal{T}, C_{2y}, \mathcal{P}, M_x\mathcal{T}, M_z\mathcal{T}, M_y, \mathcal{T}, C_{2x}, C_{2z}, C_{2y}\mathcal{T}, \mathcal{PT}, M_x, M_z, M_y\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	$p_{2b}ma'a'$ 38.11.253	$\{E, C_{2x}, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, \mathcal{P}, M_x, M_z\mathcal{T}, M_y\mathcal{T}, \mathcal{T}, C_{2x}\mathcal{T}, C_{2z}, C_{2y}, \mathcal{PT}, M_x\mathcal{T}, M_z, M_y\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_{2b}maa$ 38.13.255	$\{E, C_{2x}, C_{2z}, C_{2y}, \mathcal{P}, M_x, M_z, M_y, \mathcal{T}, C_{2x}\mathcal{T}, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, \mathcal{PT}, M_x\mathcal{T}, M_z\mathcal{T}, M_y\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2b}112/a$ 7.6.33	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, \mathcal{T}, C_{2z}, \mathcal{PT}, M_z\}$	0	0
			$\mathrm{PL}_{(010)}$	$p_{2a}2/b11$ 16.7.89	$\{E, C_{2y}, \mathcal{P}, M_y, \mathcal{T}, C_{2y}\mathcal{T}, \mathcal{PT}, M_y\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_{2b}2/m11$ 14.10.75	$\{E, C_{2x}, \mathcal{P}, M_x, \mathcal{T}, C_{2x}\mathcal{T}, \mathcal{PT}, M_x\mathcal{T}\}$	0	0
			GEN	$p_{2a}\bar{1}$ 2.4.7	$\{E, \mathcal{P}, \mathcal{T}, \mathcal{PT}\}$	0	0

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG (OG) symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
88 38.13.255 $p^1_{2b}m^1a^1a$ SI p. 49	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2x}\}$ $\{1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_x\}$ $\{1\|M_z\}$ $\{1\|M_y\}$ $\{-1\|E\}$ $\{-1\|C_{2x}\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_x\}$ $\{-1\|M_z\}$ $\{-1\|M_y\}$	$\mathrm{AX}_{[100]}$	$p_{2b}ma'a'$ 38.11.253	$\{E, C_{2x}, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, \mathcal{P}, M_x, M_z\mathcal{T}, M_y\mathcal{T}, \mathcal{T}, C_{2x}\mathcal{T}, C_{2z}, C_{2y}, \mathcal{P}\mathcal{T}, M_x\mathcal{T}, M_z, M_y\}$	0	0
			$\mathrm{AX}_{[010]}$	$p_{2b}m'aa'$ 38.10.252	$\{E, C_{2x}\mathcal{T}, C_{2z}\mathcal{T}, C_{2y}, \mathcal{P}, M_x\mathcal{T}, M_z\mathcal{T}, M_y, \mathcal{T}, C_{2x}, C_{2z}, C_{2y}\mathcal{T}, \mathcal{P}\mathcal{T}, M_x, M_z, M_y\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_{2b}m'a'a$ 38.12.254	$\{E, C_{2x}\mathcal{T}, C_{2z}, C_{2y}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}, M_z, M_y\mathcal{T}, \mathcal{T}, C_{2x}, C_{2z}\mathcal{T}, C_{2y}, \mathcal{P}\mathcal{T}, M_x, M_z\mathcal{T}, M_y\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2b}112/a$ 7.6.33	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, \mathcal{T}, C_{2z}, \mathcal{P}\mathcal{T}, M_z\}$	0	0
			$\mathrm{PL}_{(010)}$	$p_{2a}2'/b'11$ 16.6.88	$\{E, C_{2y}\mathcal{T}, \mathcal{P}, M_y\mathcal{T}, \mathcal{T}, C_{2y}, \mathcal{P}\mathcal{T}, M_y\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_{2b}2'/m'11$ 14.7.72	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}, \mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_x\}$	0	0
			GEN	$p_{2a}\bar{1}$ 2.4.7	$\{E, \mathcal{P}, \mathcal{T}, \mathcal{P}\mathcal{T}\}$	0	0
89 40.10.272 $p^{-1}_{2b}m^1a^{-1}m$ SI p. 50	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2y}\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_y\}$ $\{-1\|M_z\}$ $\{-1\|M_x\}$ $\{-1\|E\}$ $\{-1\|C_{2y}\}$ $\{1\|C_{2z}\}$ $\{1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_y\}$ $\{1\|M_z\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$p_{2b}m'a'm$ 40.11.273	$\{E, C_{2y}\mathcal{T}, C_{2z}, C_{2x}\mathcal{T}, \mathcal{P}, M_y\mathcal{T}, M_z, M_x\mathcal{T}, \mathcal{T}, C_{2y}, C_{2z}\mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_y, M_z\mathcal{T}, M_x\}$	0	0
			$\mathrm{AX}_{[010]}$	$p_{2b}mam$ 40.13.275	$\{E, C_{2y}, C_{2z}, C_{2x}, \mathcal{P}, M_y, M_z, M_x, \mathcal{T}, C_{2y}\mathcal{T}, C_{2z}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_y\mathcal{T}, M_z\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_{2b}ma'm'$ 40.12.274	$\{E, C_{2y}\mathcal{T}, C_{2z}\mathcal{T}, C_{2x}, \mathcal{P}, M_y\mathcal{T}, M_z\mathcal{T}, M_x, \mathcal{T}, C_{2y}, C_{2z}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_y, M_z, M_x\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}112/m$ 6.6.26	$\{E, C_{2z}, \mathcal{P}, M_z, \mathcal{T}, C_{2z}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	$p_{2a}2'/b'11$ 16.6.88	$\{E, C_{2y}\mathcal{T}, \mathcal{P}, M_y\mathcal{T}, \mathcal{T}, C_{2y}, \mathcal{P}\mathcal{T}, M_y\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_{2b}2_1/m11$ 15.7.82	$\{E, C_{2x}, \mathcal{P}, M_x, \mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_x\mathcal{T}\}$	0	0
			GEN	$p_{2a}\bar{1}$ 2.4.7	$\{E, \mathcal{P}, \mathcal{T}, \mathcal{P}\mathcal{T}\}$	0	0

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG (OG) symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
90 40.11.273 $p^{-1}_{2b}m^{-1}a^{1}m$ SI p. 50	$T\tau$ AFM	$\{1\|E\}$ $\{-1\|C_{2y}\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{-1\|M_y\}$ $\{1\|M_z\}$ $\{-1\|M_x\}$ $\{-1\|E\}$ $\{1\|C_{2y}\}$ $\{-1\|C_{2z}\}$ $\{1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{1\|M_y\}$ $\{-1\|M_z\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$p_{2b}m'am'$ 40.10.272	$\{E, C_{2y}, C_{2z}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}, M_y, M_z\mathcal{T}, M_x\mathcal{T}, \mathcal{T}, C_{2y}\mathcal{T}, C_{2z}, C_{2x}, \mathcal{PT}, M_y\mathcal{T}, M_z, M_x\}$	0	0
			$\mathrm{AX}_{[010]}$	$p_{2b}ma'm'$ 40.12.274	$\{E, C_{2y}\mathcal{T}, C_{2z}\mathcal{T}, C_{2x}, \mathcal{P}, M_y\mathcal{T}, M_z\mathcal{T}, M_x, \mathcal{T}, C_{2y}, C_{2z}, C_{2x}\mathcal{T}, \mathcal{PT}, M_y, M_z, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_{2b}mam$ 40.13.275	$\{E, C_{2y}, C_{2z}, C_{2x}, \mathcal{P}, M_y, M_z, M_x, \mathcal{T}, C_{2y}\mathcal{T}, C_{2z}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{PT}, M_y\mathcal{T}, M_z\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}112'/m'$ 6.7.27	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, \mathcal{T}, C_{2z}, \mathcal{PT}, M_z\}$	0	0
			$\mathrm{PL}_{(010)}$	$p_{2a}2/b11$ 16.7.89	$\{E, C_{2y}, \mathcal{P}, M_y, \mathcal{T}, C_{2y}\mathcal{T}, \mathcal{PT}, M_y\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_{2b}2_1/m11$ 15.7.82	$\{E, C_{2x}, \mathcal{P}, M_x, \mathcal{T}, C_{2x}\mathcal{T}, \mathcal{PT}, M_x\mathcal{T}\}$	0	0
			GEN	$p_{2a}\bar{1}$ 2.4.7	$\{E, \mathcal{P}, \mathcal{T}, \mathcal{PT}\}$	0	0
91 40.12.274 $p^{1}_{2b}m^{-1}a^{-1}m$ SI p. 50	$T\tau$ AFM	$\{1\|E\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2z}\}$ $\{1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{-1\|M_y\}$ $\{-1\|M_z\}$ $\{1\|M_x\}$ $\{-1\|E\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{1\|M_y\}$ $\{1\|M_z\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$p_{2b}mam$ 40.13.275	$\{E, C_{2y}, C_{2z}, C_{2x}, \mathcal{P}, M_y, M_z, M_x, \mathcal{T}, C_{2y}\mathcal{T}, C_{2z}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{PT}, M_y\mathcal{T}, M_z\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	$p_{2b}m'a'm$ 40.11.273	$\{E, C_{2y}\mathcal{T}, C_{2z}, C_{2x}\mathcal{T}, \mathcal{P}, M_y\mathcal{T}, M_z, M_x\mathcal{T}, \mathcal{T}, C_{2y}, C_{2z}\mathcal{T}, C_{2x}, \mathcal{PT}, M_y, M_z\mathcal{T}, M_x\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_{2b}m'am'$ 40.10.272	$\{E, C_{2y}, C_{2z}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}, M_y, M_z\mathcal{T}, M_x\mathcal{T}, \mathcal{T}, C_{2y}\mathcal{T}, C_{2z}, C_{2x}, \mathcal{PT}, M_y\mathcal{T}, M_z, M_x\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}112/m$ 6.6.26	$\{E, C_{2z}, \mathcal{P}, M_z, \mathcal{T}, C_{2z}\mathcal{T}, \mathcal{PT}, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	$p_{2a}2/b11$ 16.7.89	$\{E, C_{2y}, \mathcal{P}, M_y, \mathcal{T}, C_{2y}\mathcal{T}, \mathcal{PT}, M_y\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_{2b}2'_1/m'11$ 15.6.81	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}, \mathcal{T}, C_{2x}, \mathcal{PT}, M_x\}$	0	0
			GEN	$p_{2a}\bar{1}$ 2.4.7	$\{E, \mathcal{P}, \mathcal{T}, \mathcal{PT}\}$	0	0

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG (OG) symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
92 40.13.275 $p^1_{2b}m^1a^1m$ SI p. 51	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2z}\}$ $\{1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_y\}$ $\{1\|M_z\}$ $\{1\|M_x\}$ $\{-1\|E\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_y\}$ $\{-1\|M_z\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$p_{2b}ma'm'$ 40.12.274	$\{E, C_{2y}\mathcal{T}, C_{2z}\mathcal{T}, C_{2x}, \mathcal{P}, M_y\mathcal{T}, M_z\mathcal{T}, M_x, \mathcal{T}, C_{2y}, C_{2z}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_y, M_z, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	$p_{2b}m'am'$ 40.10.272	$\{E, C_{2y}, C_{2z}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}, M_y, M_z\mathcal{T}, M_x\mathcal{T}, \mathcal{T}, C_{2y}\mathcal{T}, C_{2z}, C_{2x}, \mathcal{P}\mathcal{T}, M_y\mathcal{T}, M_z, M_x\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_{2b}m'a'm$ 40.11.273	$\{E, C_{2y}\mathcal{T}, C_{2z}, C_{2x}\mathcal{T}, \mathcal{P}, M_y\mathcal{T}, M_z, M_x\mathcal{T}, \mathcal{T}, C_{2y}, C_{2z}\mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_y, M_z\mathcal{T}, M_x\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}112'/m'$ 6.7.27	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, \mathcal{T}, C_{2z}, \mathcal{P}\mathcal{T}, M_z\}$	0	0
			$\mathrm{PL}_{(010)}$	$p_{2a}2'/b'11$ 16.6.88	$\{E, C_{2y}\mathcal{T}, \mathcal{P}, M_y\mathcal{T}, \mathcal{T}, C_{2y}, \mathcal{P}\mathcal{T}, M_y\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_{2b}2'_1/m'11$ 15.6.81	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}, \mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_x\}$	0	0
			GEN	$p_{2a}\bar{1}$ 2.4.7	$\{E, \mathcal{P}, \mathcal{T}, \mathcal{P}\mathcal{T}\}$	0	0
93 41.10.285 $p^{-1}_{2b}m^1m^{-1}a$ SI p. 51	$T\tau$ AFM	$\{1\|E\}$ $\{-1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{1\|M_y\}$ $\{-1\|M_x\}$ $\{-1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{-1\|M_y\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$p_{2b}m'm'a$ 41.11.286	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}, M_z, M_y\mathcal{T}, M_x\mathcal{T}, \mathcal{T}, C_{2z}\mathcal{T}, C_{2y}, C_{2x}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, M_y, M_x\}$	0	0
			$\mathrm{AX}_{[010]}$	$p_{2b}mma$ 41.13.288	$\{E, C_{2z}, C_{2y}, C_{2x}, \mathcal{P}, M_z, M_y, M_x, \mathcal{T}, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_{2b}mm'a'$ 41.12.287	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}, M_z\mathcal{T}, M_y\mathcal{T}, M_x, \mathcal{T}, C_{2z}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z, M_y, M_x\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2b}112/a$ 7.6.33	$\{E, C_{2z}, \mathcal{P}, M_z, \mathcal{T}, C_{2z}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	$p_{2a}2'/m'11$ 14.6.71	$\{E, C_{2y}\mathcal{T}, \mathcal{P}, M_y\mathcal{T}, \mathcal{T}, C_{2y}, \mathcal{P}\mathcal{T}, M_y\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_{2b}2_1/m11$ 15.7.82	$\{E, C_{2x}, \mathcal{P}, M_x, \mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_x\mathcal{T}\}$	0	0
			GEN	$p_{2a}\bar{1}$ 2.4.7	$\{E, \mathcal{P}, \mathcal{T}, \mathcal{P}\mathcal{T}\}$	0	0

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG (OG) symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
94 41.11.286 $p^{-1}_{2b}m^{-1}m^{1}a$ SI p. 51	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$ $\{-1\|E\}$ $\{-1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{1\|M_y\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$p_{2b}m'ma'$ 41.10.285	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}, \mathcal{T}, C_{2z}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_z, M_y\mathcal{T}, M_x\}$	0	0
			$\mathrm{AX}_{[010]}$	$p_{2b}mm'a'$ 41.12.287	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}, M_z\mathcal{T}, M_y\mathcal{T}, M_x, \mathcal{T}, C_{2z}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z, M_y, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_{2b}mma$ 41.13.288	$\{E, C_{2z}, C_{2y}, C_{2x}, \mathcal{P}, M_z, M_y, M_x, \mathcal{T}, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2b}112/a$ 7.6.33	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, \mathcal{T}, C_{2z}, \mathcal{P}\mathcal{T}, M_z\}$	0	0
			$\mathrm{PL}_{(010)}$	$p_{2a}2/m11$ 14.9.74	$\{E, C_{2y}, \mathcal{P}, M_y, \mathcal{T}, C_{2y}\mathcal{T}, \mathcal{P}\mathcal{T}, M_y\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_{2b}2_1/m11$ 15.7.82	$\{E, C_{2x}, \mathcal{P}, M_x, \mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_x\mathcal{T}\}$	0	0
			GEN	$p_{2a}\bar{1}$ 2.4.7	$\{E, \mathcal{P}, \mathcal{T}, \mathcal{P}\mathcal{T}\}$	0	0
95 41.12.287 $p^{1}_{2b}m^{-1}m^{-1}a$ SI p. 52	$T\tau$ AFM	$\{1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{-1\|M_y\}$ $\{1\|M_x\}$ $\{-1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{1\|M_y\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$p_{2b}mma$ 41.13.288	$\{E, C_{2z}, C_{2y}, C_{2x}, \mathcal{P}, M_z, M_y, M_x, \mathcal{T}, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	$p_{2b}m'm'a$ 41.11.286	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}, M_z, M_y\mathcal{T}, M_x\mathcal{T}, \mathcal{T}, C_{2z}\mathcal{T}, C_{2y}, C_{2x}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, M_y, M_x\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_{2b}m'ma'$ 41.10.285	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}, \mathcal{T}, C_{2z}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_z, M_y\mathcal{T}, M_x\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2b}112/a$ 7.6.33	$\{E, C_{2z}, \mathcal{P}, M_z, \mathcal{T}, C_{2z}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	$p_{2a}2/m11$ 14.9.74	$\{E, C_{2y}, \mathcal{P}, M_y, \mathcal{T}, C_{2y}\mathcal{T}, \mathcal{P}\mathcal{T}, M_y\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_{2b}2'_1/m'11$ 15.6.81	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}, \mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_x\}$	0	0
			GEN	$p_{2a}\bar{1}$ 2.4.7	$\{E, \mathcal{P}, \mathcal{T}, \mathcal{P}\mathcal{T}\}$	0	0

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG (OG) symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
96 41.13.288 $p^1_{2b}m^1m^1a$ SI p. 52	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{1\|M_y\}$ $\{1\|M_x\}$ $\{-1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$p_{2b}mm'a'$ 41.12.287	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}, M_z\mathcal{T}, M_y\mathcal{T}, M_x, \mathcal{T}, C_{2z}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z, M_y, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	$p_{2b}m'ma'$ 41.10.285	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}, \mathcal{T}, C_{2z}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_z, M_y\mathcal{T}, M_x\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_{2b}m'm'a$ 41.11.286	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}, M_z, M_y\mathcal{T}, M_x\mathcal{T}, \mathcal{T}, C_{2z}\mathcal{T}, C_{2y}, C_{2x}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, M_y, M_x\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2b}112/a$ 7.6.33	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, \mathcal{T}, C_{2z}, \mathcal{P}\mathcal{T}, M_z\}$	0	0
			$\mathrm{PL}_{(010)}$	$p_{2a}2'/m'11$ 14.6.71	$\{E, C_{2y}\mathcal{T}, \mathcal{P}, M_y\mathcal{T}, \mathcal{T}, C_{2y}, \mathcal{P}\mathcal{T}, M_y\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_{2b}2'_1/m'11$ 15.6.81	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}, \mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_x\}$	0	0
			GEN	$p_{2a}\bar{1}$ 2.4.7	$\{E, \mathcal{P}, \mathcal{T}, \mathcal{P}\mathcal{T}\}$	0	0
97 47.8.337 $c^1_pm^1m^1m$ SI p. 52	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{1\|M_y\}$ $\{1\|M_x\}$ $\{-1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$c_pmm'm'$ 47.11.340	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}, M_z\mathcal{T}, M_y\mathcal{T}, M_x, \mathcal{T}, C_{2z}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z, M_y, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	$c_pmm'm'$ 47.11.340	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}, \mathcal{T}, C_{2z}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_z, M_y\mathcal{T}, M_x\}$	0	0
			$\mathrm{AX}_{[001]}$	$c_pm'm'm$ 47.10.339	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}, M_z, M_y\mathcal{T}, M_x\mathcal{T}, \mathcal{T}, C_{2z}\mathcal{T}, C_{2y}, C_{2x}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, M_y, M_x\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}112'/m'$ 6.7.27	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, \mathcal{T}, C_{2z}, \mathcal{P}\mathcal{T}, M_z\}$	0	0
			$\mathrm{PL}_{(010)}$	$c_p2'/m'11$ 18.8.102	$\{E, C_{2y}\mathcal{T}, \mathcal{P}, M_y\mathcal{T}, \mathcal{T}, C_{2y}, \mathcal{P}\mathcal{T}, M_y\}$	0	0
			$\mathrm{PL}_{(100)}$	$c_p2'/m'11$ 18.8.102	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}, \mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_x\}$	0	0
			GEN	$p_{2a}\bar{1}$ 2.4.7	$\{E, \mathcal{P}, \mathcal{T}, \mathcal{P}\mathcal{T}\}$	0	0

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG (OG) symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
98 47.9.338 $c_p^{-1}m^{-1}m^{-1}m$ SI p. 53	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$ $\{-1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{1\|M_y\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$c_pmm'm$ 47.13.342	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_z, M_y, M_x\mathcal{T}, \mathcal{T}, C_{2z}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	0
			$\mathrm{AX}_{[010]}$	$c_pmm'm$ 47.13.342	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z, M_y\mathcal{T}, M_x, \mathcal{T}, C_{2z}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	c_pmmm' 47.12.341	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, M_y, M_x, \mathcal{T}, C_{2z}\mathcal{T}, C_{2y}, C_{2x}, \mathcal{P}, M_z, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}112/m$ 6.6.26	$\{E, C_{2z}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z, \mathcal{T}, C_{2z}, \mathcal{P}, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	$c_p2'/m11$ 18.9.103	$\{E, C_{2y}\mathcal{T}, \mathcal{P}\mathcal{T}, M_y, \mathcal{T}, C_{2y}, \mathcal{P}, M_y\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(100)}$	$c_p2'/m11$ 18.9.103	$\{E, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_x, \mathcal{T}, C_{2x}, \mathcal{P}, M_x\mathcal{T}\}$	0	0
			GEN	$p_{2a}\bar{1}$ 2.4.7	$\{E, \mathcal{P}\mathcal{T}, \mathcal{T}, \mathcal{P}\}$	0	0
99 47.10.339 $c_p^{-1}m^{-1}m^1m$ SI p. 53	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$ $\{-1\|E\}$ $\{-1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{1\|M_y\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$c_pmm'm'$ 47.11.340	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}, \mathcal{T}, C_{2z}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_z, M_y\mathcal{T}, M_x\}$	0	0
			$\mathrm{AX}_{[010]}$	$c_pmm'm'$ 47.11.340	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}, M_z\mathcal{T}, M_y\mathcal{T}, M_x, \mathcal{T}, C_{2z}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z, M_y, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	c_pmmm 47.8.337	$\{E, C_{2z}, C_{2y}, C_{2x}, \mathcal{P}, M_z, M_y, M_x, \mathcal{T}, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}112'/m'$ 6.7.27	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, \mathcal{T}, C_{2z}, \mathcal{P}\mathcal{T}, M_z\}$	0	0
			$\mathrm{PL}_{(010)}$	$c_p2/m11$ 18.6.100	$\{E, C_{2y}, \mathcal{P}, M_y, \mathcal{T}, C_{2y}\mathcal{T}, \mathcal{P}\mathcal{T}, M_y\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(100)}$	$c_p2/m11$ 18.6.100	$\{E, C_{2x}, \mathcal{P}, M_x, \mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_x\mathcal{T}\}$	0	0
			GEN	$p_{2a}\bar{1}$ 2.4.7	$\{E, \mathcal{P}, \mathcal{T}, \mathcal{P}\mathcal{T}\}$	0	0

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG (OG) symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
100 47.11.340 $c_p^1m^{-1}m^{-1}m$ SI p. 54	$T\tau$ AFM	$\{1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{-1\|M_y\}$ $\{1\|M_x\}$ $\{-1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{1\|M_y\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	c_pmmm 47.8.337	$\{E, C_{2z}, C_{2y}, C_{2x}, \mathcal{P}, M_z, M_y, M_x, \mathcal{T}, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{PT}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	$c_pm'm'm$ 47.10.339	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}, M_z, M_y\mathcal{T}, M_x\mathcal{T}, \mathcal{T}, C_{2z}\mathcal{T}, C_{2y}, C_{2x}, \mathcal{PT}, M_z\mathcal{T}, M_y, M_x\}$	0	0
			$\mathrm{AX}_{[001]}$	$c_pmm'm'$ 47.11.340	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}, \mathcal{T}, C_{2z}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{PT}, M_z, M_y\mathcal{T}, M_x\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}112/m$ 6.6.26	$\{E, C_{2z}, \mathcal{P}, M_z, \mathcal{T}, C_{2z}\mathcal{T}, \mathcal{PT}, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	$c_p2/m11$ 18.6.100	$\{E, C_{2y}, \mathcal{P}, M_y, \mathcal{T}, C_{2y}\mathcal{T}, \mathcal{PT}, M_y\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(100)}$	$c_p2'/m'11$ 18.8.102	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}, \mathcal{T}, C_{2x}, \mathcal{PT}, M_x\}$	0	0
			GEN	$p_{2a}\bar{1}$ 2.4.7	$\{E, \mathcal{P}, \mathcal{T}, \mathcal{PT}\}$	0	0
101 47.12.341 $c_p^1m^1m^{-1}m$ SI p. 54	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{1\|M_y\}$ $\{1\|M_x\}$ $\{-1\|E\}$ $\{-1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$c_pmm'm$ 47.13.342	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{PT}, M_z, M_y\mathcal{T}, M_x, \mathcal{T}, C_{2z}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	$c_pmm'm$ 47.13.342	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{PT}, M_z, M_y, M_x\mathcal{T}, \mathcal{T}, C_{2z}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	0
			$\mathrm{AX}_{[001]}$	$c_pm'm'm'$ 47.9.338	$\{E, C_{2z}, C_{2y}, C_{2x}, \mathcal{PT}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\mathcal{T}, \mathcal{T}, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}, M_z, M_y, M_x\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}112/m$ 6.6.26	$\{E, C_{2z}\mathcal{T}, \mathcal{PT}, M_z, \mathcal{T}, C_{2z}, \mathcal{P}, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	$c_p2/m'11$ 18.7.101	$\{E, C_{2y}, \mathcal{PT}, M_y\mathcal{T}, \mathcal{T}, C_{2y}\mathcal{T}, \mathcal{P}, M_y\}$	0	0
			$\mathrm{PL}_{(100)}$	$c_p2/m'11$ 18.7.101	$\{E, C_{2x}, \mathcal{PT}, M_x\mathcal{T}, \mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}, M_x\}$	0	0
			GEN	$p_{2a}\bar{1}$ 2.4.7	$\{E, \mathcal{PT}, \mathcal{T}, \mathcal{P}\}$	0	0

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG (OG) symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
102 47.13.342 $c_p^1m^{-1}m^1m$ SI p. 54	$T\tau$ AFM	$\{1\|E\}$ $\{-1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{-1\|M_y\}$ $\{1\|M_x\}$ $\{-1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{1\|M_y\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	c_pmmm' 47.12.341	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, M_y, M_x, \mathcal{T}, C_{2z}\mathcal{T}, C_{2y}, C_{2x}, \mathcal{P}, M_z, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	$c_pm'm'm'$ 47.9.338	$\{E, C_{2z}, C_{2y}, C_{2x}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\mathcal{T}, \mathcal{T}, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}, M_z, M_y, M_x\}$	0	0
			$\mathrm{AX}_{[001]}$	$c_pmm'm$ 47.13.342	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_z, M_y, M_x\mathcal{T}, \mathcal{T}, C_{2z}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}112'/m'$ 6.7.27	$\{E, C_{2z}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, \mathcal{T}, C_{2z}\mathcal{T}, \mathcal{P}, M_z\}$	0	0
			$\mathrm{PL}_{(010)}$	$c_p2'/m11$ 18.9.103	$\{E, C_{2y}\mathcal{T}, \mathcal{P}\mathcal{T}, M_y, \mathcal{T}, C_{2y}, \mathcal{P}, M_y\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(100)}$	$c_p2/m'11$ 18.7.101	$\{E, C_{2x}, \mathcal{P}\mathcal{T}, M_x\mathcal{T}, \mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}, M_x\}$	0	0
			GEN	$p_{2a}\bar{1}$ 2.4.7	$\{E, \mathcal{P}\mathcal{T}, \mathcal{T}, \mathcal{P}\}$	0	0
103 48.8.350 $c_p^{-1}m^{-1}m^{-1}e$ SI p. 55	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$ $\{-1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{1\|M_y\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$c_pmm'e'$ 48.10.352	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_z, M_y, M_x\mathcal{T}, \mathcal{T}, C_{2z}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	0
			$\mathrm{AX}_{[010]}$	$c_pm'me'$ 48.9.351	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z, M_y\mathcal{T}, M_x, \mathcal{T}, C_{2z}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	c_pmme' 48.11.353	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, M_y, M_x, \mathcal{T}, C_{2z}\mathcal{T}, C_{2y}, C_{2x}, \mathcal{P}, M_z, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2b}112/a$ 7.6.33	$\{E, C_{2z}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z, \mathcal{T}, C_{2z}, \mathcal{P}, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	$c_p2/m11$ 18.6.100	$\{E, C_{2y}\mathcal{T}, \mathcal{P}\mathcal{T}, M_y, \mathcal{T}, C_{2y}, \mathcal{P}, M_y\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(100)}$	$c_p2'/m11$ 18.9.103	$\{E, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_x, \mathcal{T}, C_{2x}, \mathcal{P}, M_x\mathcal{T}\}$	0	0
			GEN	$p_{2a}\bar{1}$ 2.4.7	$\{E, \mathcal{P}\mathcal{T}, \mathcal{T}, \mathcal{P}\}$	0	0

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG (OG) symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
104 48.9.351 $c_p^{-1}m^1m^{-1}e$ SI p. 55	$T\tau$ AFM	$\{1\|E\}$ $\{-1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{1\|M_y\}$ $\{-1\|M_x\}$ $\{-1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{-1\|M_y\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$c_pm'm'e'$ 48.8.350	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}, M_z, M_y\mathcal{T}, M_x\mathcal{T}, \mathcal{T}, C_{2z}\mathcal{T}, C_{2y}, C_{2x}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, M_y, M_x\}$	0	0
			$\mathrm{AX}_{[010]}$	c_pmme' 48.11.353	$\{E, C_{2z}, C_{2y}, C_{2x}, \mathcal{P}, M_z, M_y, M_x, \mathcal{T}, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$c_pmm'e'$ 48.10.352	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}, M_z\mathcal{T}, M_y\mathcal{T}, M_x, \mathcal{T}, C_{2z}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z, M_y, M_x\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2b}112/a$ 7.6.33	$\{E, C_{2z}, \mathcal{P}, M_z, \mathcal{T}, C_{2z}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	$c_p2/m'11$ 18.7.101	$\{E, C_{2y}\mathcal{T}, \mathcal{P}, M_y\mathcal{T}, \mathcal{T}, C_{2y}, \mathcal{P}\mathcal{T}, M_y\}$	0	0
			$\mathrm{PL}_{(100)}$	$c_p2/m11$ 18.6.100	$\{E, C_{2x}, \mathcal{P}, M_x, \mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_x\mathcal{T}\}$	0	0
			GEN	$p_{2a}\bar{1}$ 2.4.7	$\{E, \mathcal{P}, \mathcal{T}, \mathcal{P}\mathcal{T}\}$	0	0
105 48.10.352 $c_p^1m^{-1}m^{-1}e$ SI p. 55	$T\tau$ AFM	$\{1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{-1\|M_y\}$ $\{1\|M_x\}$ $\{-1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{1\|M_y\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	c_pmme' 48.11.353	$\{E, C_{2z}, C_{2y}, C_{2x}, \mathcal{P}, M_z, M_y, M_x, \mathcal{T}, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	$c_pm'm'e'$ 48.8.350	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}, M_z, M_y\mathcal{T}, M_x\mathcal{T}, \mathcal{T}, C_{2z}\mathcal{T}, C_{2y}, C_{2x}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, M_y, M_x\}$	0	0
			$\mathrm{AX}_{[001]}$	$c_pm'me'$ 48.9.351	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}, \mathcal{T}, C_{2z}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_z, M_y\mathcal{T}, M_x\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2b}112/a$ 7.6.33	$\{E, C_{2z}, \mathcal{P}, M_z, \mathcal{T}, C_{2z}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	$c_p2'/m11$ 18.9.103	$\{E, C_{2y}, \mathcal{P}, M_y, \mathcal{T}, C_{2y}\mathcal{T}, \mathcal{P}\mathcal{T}, M_y\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(100)}$	$c_p2'/m'11$ 18.8.102	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}, \mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_x\}$	0	0
			GEN	$p_{2a}\bar{1}$ 2.4.7	$\{E, \mathcal{P}, \mathcal{T}, \mathcal{P}\mathcal{T}\}$	0	0

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG (OG) symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
106 48.11.353 $c_p^1 m^1 m^{-1} e$ SI p. 56	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{1\|M_y\}$ $\{1\|M_x\}$ $\{-1\|E\}$ $\{-1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$c_p m' m e'$ 48.9.351	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z, M_y\mathcal{T}, M_x, \mathcal{T}, C_{2z}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	$c_p m m' e'$ 48.10.352	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_z, M_y, M_x\mathcal{T}, \mathcal{T}, C_{2z}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	0
			$\mathrm{AX}_{[001]}$	$c_p m' m' e'$ 48.8.350	$\{E, C_{2z}, C_{2y}, C_{2x}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\mathcal{T}, \mathcal{T}, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}, M_z, M_y, M_x\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2b}112/a$ 7.6.33	$\{E, C_{2z}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z, \mathcal{T}, C_{2z}, \mathcal{P}, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	$c_p 2'/m'11$ 18.8.102	$\{E, C_{2y}, \mathcal{P}\mathcal{T}, M_y\mathcal{T}, \mathcal{T}, C_{2y}\mathcal{T}, \mathcal{P}, M_y\}$	0	0
			$\mathrm{PL}_{(100)}$	$c_p 2/m'11$ 18.7.101	$\{E, C_{2x}, \mathcal{P}\mathcal{T}, M_x\mathcal{T}, \mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}, M_x\}$	0	0
			GEN	$p_{2a}\bar{1}$ 2.4.7	$\{E, \mathcal{P}\mathcal{T}, \mathcal{T}, \mathcal{P}\}$	0	0
107 49.4.357 $p_p^1 4$ SI p. 56	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{4z}\}$ $\{1\|C_{4z}^{-1}\}$ $\{-1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{4z}\}$ $\{-1\|C_{4z}^{-1}\}$	$\mathrm{AX}_{[001]}$	$p_p 4$ 49.4.357	$\{E, C_{2z}, C_{4z}, C_{4z}^{-1}, \mathcal{T}, C_{2z}\mathcal{T}, C_{4z}\mathcal{T}, C_{4z}^{-1}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}112$ 3.4.11	$\{E, C_{2z}\mathcal{T}, \mathcal{T}, C_{2z}\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0
108 (S1) 50.4.361 $p_p^1 \bar{4}$ SI p. 56	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|\bar{C}_{4z}\}$ $\{1\|\bar{C}_{4z}^{-1}\}$ $\{-1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|\bar{C}_{4z}\}$ $\{-1\|\bar{C}_{4z}^{-1}\}$	$\mathrm{AX}_{[001]}$	$p_p \bar{4}$ 50.4.361	$\{E, C_{2z}, \bar{C}_{4z}, \bar{C}_{4z}^{-1}, \mathcal{T}, C_{2z}\mathcal{T}, \bar{C}_{4z}\mathcal{T}, \bar{C}_{4z}^{-1}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}112$ 3.4.11	$\{E, C_{2z}\mathcal{T}, \mathcal{T}, C_{2z}\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG (OG) symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
109 51.6.367 $p^1_p4/^{-1}m$ SI p. 56	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{4z}\}$ $\{1\|C^{-1}_{4z}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{-1\|\bar{C}_{4z}\}$ $\{-1\|\bar{C}^{-1}_{4z}\}$ $\{-1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{4z}\}$ $\{-1\|C^{-1}_{4z}\}$	$\mathrm{AX}_{[001]}$	p_p4/m' 51.6.367	$\{E, C_{2z}, C_{4z}, C^{-1}_{4z}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, \bar{C}_{4z}\mathcal{T}, \bar{C}^{-1}_{4z}\mathcal{T}, \mathcal{T}, C_{2z}\mathcal{T}, C_{4z}\mathcal{T}, C^{-1}_{4z}\mathcal{T}, \mathcal{P}, M_z, \bar{C}_{4z}, \bar{C}^{-1}_{4z}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}112/m$ 6.6.26	$\{E, C_{2z}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z, \mathcal{T}, C_{2z}, \mathcal{P}, M_z\mathcal{T}\}$	0	0
		$\{1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{1\|\bar{C}_{4z}\}$ $\{1\|\bar{C}^{-1}_{4z}\}$	GEN	$p_{2a}\bar{1}$ 2.4.7	$\{E, \mathcal{P}\mathcal{T}, \mathcal{T}, \mathcal{P}\}$	0	0
110 51.7.368 $p^1_p4/^1m$ SI p. 57	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{4z}\}$ $\{1\|C^{-1}_{4z}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{1\|\bar{C}_{4z}\}$ $\{1\|\bar{C}^{-1}_{4z}\}$ $\{-1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{4z}\}$ $\{-1\|C^{-1}_{4z}\}$	$\mathrm{AX}_{[001]}$	p_p4/m 51.7.368	$\{E, C_{2z}, C_{4z}, C^{-1}_{4z}, \mathcal{P}, M_z, \bar{C}_{4z}, \bar{C}^{-1}_{4z}, \mathcal{T}, C_{2z}\mathcal{T}, C_{4z}\mathcal{T}, C^{-1}_{4z}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, \bar{C}_{4z}\mathcal{T}, \bar{C}^{-1}_{4z}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}112'/m'$ 6.7.27	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, \mathcal{T}, C_{2z}, \mathcal{P}\mathcal{T}, M_z\}$	0	0
		$\{-1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{-1\|\bar{C}_{4z}\}$ $\{-1\|\bar{C}^{-1}_{4z}\}$	GEN	$p_{2a}\bar{1}$ 2.4.7	$\{E, \mathcal{P}, \mathcal{T}, \mathcal{P}\mathcal{T}\}$	0	0

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG (OG) symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
111 53.6.379 $p^1_p4^{-1}2^{-1}2$ SI p. 57	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{4z}\}$ $\{1\|C^{-1}_{4z}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{-1\|C_{2xy}\}$ $\{-1\|C_{2\bar{x}y}\}$ $\{-1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{4z}\}$ $\{-1\|C^{-1}_{4z}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{1\|C_{2xy}\}$ $\{1\|C_{2\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	p_c222 19.6.109	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{T}, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\}$	0	0
			$\mathrm{AX}_{[010]}$	p_c222 19.6.109	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{T}, C_{2z}, C_{2y}, C_{2x}\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[110]}$	$c_p2'22'$ 22.7.128	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}, \mathcal{T}, C_{2z}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[1\bar{1}0]}$	$c_p2'22'$ 22.7.128	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}, \mathcal{T}, C_{2z}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}\}$	0	0
			$\mathrm{AX}_{[001]}$	p_p422 53.7.380	$\{E, C_{2z}, C_{4z}, C^{-1}_{4z}, C_{2y}, C_{2x}, C_{2xy}, C_{2\bar{x}y}, \mathcal{T}, C_{2z}\mathcal{T}, C_{4z}\mathcal{T}, C^{-1}_{4z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}112$ 3.4.11	$\{E, C_{2z}\mathcal{T}, \mathcal{T}, C_{2z}\}$	0	0
			$\mathrm{PL}_{(010)}$	p_c211 8.5.38	$\{E, C_{2y}, \mathcal{T}, C_{2y}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(100)}$	p_c211 8.5.38	$\{E, C_{2x}, \mathcal{T}, C_{2x}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(110)}$	c_p211 10.4.48	$\{E, C_{2xy}, \mathcal{T}, C_{2xy}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(1\bar{1}0)}$	c_p211 10.4.48	$\{E, C_{2\bar{x}y}, \mathcal{T}, C_{2\bar{x}y}\mathcal{T}\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0
112 53.7.380 $p^1_p4^12^12$ SI p. 58	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{4z}\}$ $\{1\|C^{-1}_{4z}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{1\|C_{2xy}\}$ $\{1\|C_{2\bar{x}y}\}$ $\{-1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{4z}\}$ $\{-1\|C^{-1}_{4z}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{-1\|C_{2xy}\}$ $\{-1\|C_{2\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	p_c222 19.6.109	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{T}, C_{2z}, C_{2y}, C_{2x}\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	p_c222 19.6.109	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{T}, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\}$	0	0
			$\mathrm{AX}_{[110]}$	$c_p2'22'$ 22.7.128	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}, \mathcal{T}, C_{2z}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}\}$	0	0
			$\mathrm{AX}_{[1\bar{1}0]}$	$c_p2'22'$ 22.7.128	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}, \mathcal{T}, C_{2z}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_p42'2'$ 53.6.379	$\{E, C_{2z}, C_{4z}, C^{-1}_{4z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}\mathcal{T}, \mathcal{T}, C_{2z}\mathcal{T}, C_{4z}\mathcal{T}, C^{-1}_{4z}\mathcal{T}, C_{2y}, C_{2x}, C_{2xy}, C_{2\bar{x}y}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}112$ 3.4.11	$\{E, C_{2z}\mathcal{T}, \mathcal{T}, C_{2z}\}$	0	0
			$\mathrm{PL}_{(010)}$	p_c211 8.5.38	$\{E, C_{2y}\mathcal{T}, \mathcal{T}, C_{2y}\}$	0	0
			$\mathrm{PL}_{(100)}$	p_c211 8.5.38	$\{E, C_{2x}\mathcal{T}, \mathcal{T}, C_{2x}\}$	0	0
			$\mathrm{PL}_{(110)}$	$c_p2'11$ 10.5.49	$\{E, C_{2xy}\mathcal{T}, \mathcal{T}, C_{2xy}\}$	0	0
			$\mathrm{PL}_{(1\bar{1}0)}$	$c_p2'11$ 10.5.49	$\{E, C_{2\bar{x}y}\mathcal{T}, \mathcal{T}, C_{2\bar{x}y}\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG (OG) symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
113 55.6.391 $p^1_p 4^{-1} m^{-1} m$ SI p. 58	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{4z}\}$ $\{1\|C^{-1}_{4z}\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$ $\{-1\|M_{xy}\}$ $\{-1\|M_{\bar{x}y}\}$ $\{-1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{4z}\}$ $\{-1\|C^{-1}_{4z}\}$ $\{1\|M_y\}$ $\{1\|M_x\}$ $\{1\|M_{xy}\}$ $\{1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$p_c mm2$ 23.6.134	$\{E, C_{2z}\mathcal{T}, M_y, M_x\mathcal{T}, \mathcal{T}, C_{2z}, M_y\mathcal{T}, M_x\}$	0	0
			$\mathrm{AX}_{[010]}$	$p_c mm2$ 23.6.134	$\{E, C_{2z}\mathcal{T}, M_y\mathcal{T}, M_x, \mathcal{T}, C_{2z}, M_y, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[110]}$	$c_p mm'2'$ 26.7.153	$\{E, C_{2z}\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}, \mathcal{T}, C_{2z}, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[1\bar{1}0]}$	$c_p mm'2'$ 26.7.153	$\{E, C_{2z}\mathcal{T}, M_{xy}, M_{\bar{x}y}\mathcal{T}, \mathcal{T}, C_{2z}, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_p 4mm$ 55.7.392	$\{E, C_{2z}, C_{4z}, C^{-1}_{4z}, M_y, M_x, M_{xy}, M_{\bar{x}y}, \mathcal{T}, C_{2z}\mathcal{T}, C_{4z}\mathcal{T}, C^{-1}_{4z}\mathcal{T}, M_y\mathcal{T}, M_x\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}112$ 3.4.11	$\{E, C_{2z}\mathcal{T}, \mathcal{T}, C_{2z}\}$	0	0
			$\mathrm{PL}_{(010)}$	$p_c m11$ 11.7.56	$\{E, M_y, \mathcal{T}, M_y\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_c m11$ 11.7.56	$\{E, M_x, \mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(110)}$	$c_p m11$ 13.4.64	$\{E, M_{xy}, \mathcal{T}, M_{xy}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(1\bar{1}0)}$	$c_p m11$ 13.4.64	$\{E, M_{\bar{x}y}, \mathcal{T}, M_{\bar{x}y}\mathcal{T}\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0
114 55.7.392 $p^1_p 4^1 m^1 m$ SI p. 58	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{4z}\}$ $\{1\|C^{-1}_{4z}\}$ $\{1\|M_y\}$ $\{1\|M_x\}$ $\{1\|M_{xy}\}$ $\{1\|M_{\bar{x}y}\}$ $\{-1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{4z}\}$ $\{-1\|C^{-1}_{4z}\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$ $\{-1\|M_{xy}\}$ $\{-1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$p_c mm2$ 23.6.134	$\{E, C_{2z}\mathcal{T}, M_y\mathcal{T}, M_x, \mathcal{T}, C_{2z}, M_y, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	$p_c mm2$ 23.6.134	$\{E, C_{2z}\mathcal{T}, M_y, M_x\mathcal{T}, \mathcal{T}, C_{2z}, M_y\mathcal{T}, M_x\}$	0	0
			$\mathrm{AX}_{[110]}$	$c_p mm'2'$ 26.7.153	$\{E, C_{2z}\mathcal{T}, M_{xy}, M_{\bar{x}y}\mathcal{T}, \mathcal{T}, C_{2z}, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	0	0
			$\mathrm{AX}_{[1\bar{1}0]}$	$c_p mm'2'$ 26.7.153	$\{E, C_{2z}\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}, \mathcal{T}, C_{2z}, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_p 4m'm'$ 55.6.391	$\{E, C_{2z}, C_{4z}, C^{-1}_{4z}, M_y\mathcal{T}, M_x\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}\mathcal{T}, \mathcal{T}, C_{2z}\mathcal{T}, C_{4z}\mathcal{T}, C^{-1}_{4z}\mathcal{T}, M_y, M_x, M_{xy}, M_{\bar{x}y}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}112$ 3.4.11	$\{E, C_{2z}\mathcal{T}, \mathcal{T}, C_{2z}\}$	0	0
			$\mathrm{PL}_{(010)}$	$p_c m11$ 11.7.56	$\{E, M_y\mathcal{T}, \mathcal{T}, M_y\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_c m11$ 11.7.56	$\{E, M_x\mathcal{T}, \mathcal{T}, M_x\}$	0	0
			$\mathrm{PL}_{(110)}$	$c_p m'11$ 13.5.65	$\{E, M_{xy}\mathcal{T}, \mathcal{T}, M_{xy}\}$	0	0
			$\mathrm{PL}_{(1\bar{1}0)}$	$c_p m'11$ 13.5.65	$\{E, M_{\bar{x}y}\mathcal{T}, \mathcal{T}, M_{\bar{x}y}\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG (OG) symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
115 57.6.403 $p^1_p\bar{4}^1 2^1 m$ SI p. 59	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|\bar{C}_{4z}\}$ $\{1\|\bar{C}^{-1}_{4z}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{1\|M_{xy}\}$ $\{1\|M_{\bar{x}y}\}$ $\{-1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|\bar{C}_{4z}\}$ $\{-1\|\bar{C}^{-1}_{4z}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{-1\|M_{xy}\}$ $\{-1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	p_c222 19.6.109	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{T}, C_{2z}, C_{2y}, C_{2x}\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	p_c222 19.6.109	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{T}, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\}$	0	0
			$\mathrm{AX}_{[110]}$	$c_pmm'2'$ 26.7.153	$\{E, C_{2z}\mathcal{T}, M_{xy}, M_{\bar{x}y}\mathcal{T}, \mathcal{T}, C_{2z}, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	0	0
			$\mathrm{AX}_{[1\bar{1}0]}$	$c_pmm'2'$ 26.7.153	$\{E, C_{2z}\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}, \mathcal{T}, C_{2z}, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_p\bar{4}2'm'$ 57.7.404	$\{E, C_{2z}, \bar{C}_{4z}, \bar{C}^{-1}_{4z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}\mathcal{T}, \mathcal{T}, C_{2z}\mathcal{T}, \bar{C}_{4z}\mathcal{T}, \bar{C}^{-1}_{4z}\mathcal{T}, C_{2y}, C_{2x}, M_{xy}, M_{\bar{x}y}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}112$ 3.4.11	$\{E, C_{2z}\mathcal{T}, \mathcal{T}, C_{2z}\}$	0	0
			$\mathrm{PL}_{(010)}$	p_c211 8.5.38	$\{E, C_{2y}\mathcal{T}, \mathcal{T}, C_{2y}\}$	0	0
			$\mathrm{PL}_{(100)}$	p_c211 8.5.38	$\{E, C_{2x}\mathcal{T}, \mathcal{T}, C_{2x}\}$	0	0
			$\mathrm{PL}_{(110)}$	$c_pm'11$ 13.5.65	$\{E, M_{xy}\mathcal{T}, \mathcal{T}, M_{xy}\}$	0	0
			$\mathrm{PL}_{(1\bar{1}0)}$	$c_pm'11$ 13.5.65	$\{E, M_{\bar{x}y}\mathcal{T}, \mathcal{T}, M_{\bar{x}y}\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0
116 57.7.404 $p^1_p\bar{4}^{-1}2^{-1}m$ SI p. 59	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|\bar{C}_{4z}\}$ $\{1\|\bar{C}^{-1}_{4z}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{-1\|M_{xy}\}$ $\{-1\|M_{\bar{x}y}\}$ $\{-1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|\bar{C}_{4z}\}$ $\{-1\|\bar{C}^{-1}_{4z}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{1\|M_{xy}\}$ $\{1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	p_c222 19.6.109	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{T}, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\}$	0	0
			$\mathrm{AX}_{[010]}$	p_c222 19.6.109	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{T}, C_{2z}, C_{2y}, C_{2x}\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[110]}$	$c_pmm'2'$ 26.7.153	$\{E, C_{2z}\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}, \mathcal{T}, C_{2z}, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[1\bar{1}0]}$	$c_pmm'2'$ 26.7.153	$\{E, C_{2z}\mathcal{T}, M_{xy}, M_{\bar{x}y}\mathcal{T}, \mathcal{T}, C_{2z}, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	0	0
			$\mathrm{AX}_{[001]}$	p_p42m 57.6.403	$\{E, C_{2z}, \bar{C}_{4z}, \bar{C}^{-1}_{4z}, C_{2y}, C_{2x}, M_{xy}, M_{\bar{x}y}, \mathcal{T}, C_{2z}\mathcal{T}, \bar{C}_{4z}\mathcal{T}, \bar{C}^{-1}_{4z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}112$ 3.4.11	$\{E, C_{2z}\mathcal{T}, \mathcal{T}, C_{2z}\}$	0	0
			$\mathrm{PL}_{(010)}$	p_c211 8.5.38	$\{E, C_{2y}, \mathcal{T}, C_{2y}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(100)}$	p_c211 8.5.38	$\{E, C_{2x}, \mathcal{T}, C_{2x}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(110)}$	c_pm11 13.4.64	$\{E, M_{xy}, \mathcal{T}, M_{xy}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(1\bar{1}0)}$	c_pm11 13.4.64	$\{E, M_{\bar{x}y}, \mathcal{T}, M_{\bar{x}y}\mathcal{T}\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG (OG) symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
117 59.6.415 $p_p^1\bar{4}^1m^12$ SI p. 59	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|\bar{C}_{4z}\}$ $\{1\|\bar{C}_{4z}^{-1}\}$ $\{1\|M_y\}$ $\{1\|M_x\}$ $\{1\|C_{2xy}\}$ $\{1\|C_{2\bar{x}y}\}$ $\{-1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|\bar{C}_{4z}\}$ $\{-1\|\bar{C}_{4z}^{-1}\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$ $\{-1\|C_{2xy}\}$ $\{-1\|C_{2\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	p_cmm2 23.6.134	$\{E, C_{2z}\mathcal{T}, M_y\mathcal{T}, M_x, \mathcal{T}, C_{2z}, M_y, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	p_cmm2 23.6.134	$\{E, C_{2z}\mathcal{T}, M_y, M_x\mathcal{T}, \mathcal{T}, C_{2z}, M_y\mathcal{T}, M_x\}$	0	0
			$\mathrm{AX}_{[110]}$	$c_p2'22'$ 22.7.128	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}, \mathcal{T}, C_{2z}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}\}$	0	0
			$\mathrm{AX}_{[1\bar{1}0]}$	$c_p2'22'$ 22.7.128	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}, \mathcal{T}, C_{2z}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_p\bar{4}m'2'$ 59.7.416	$\{E, C_{2z}, \bar{C}_{4z}, \bar{C}_{4z}^{-1}, M_y\mathcal{T}, M_x\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}\mathcal{T}, \mathcal{T}, C_{2z}\mathcal{T}, \bar{C}_{4z}\mathcal{T}, \bar{C}_{4z}^{-1}\mathcal{T}, M_y, M_x, C_{2xy}, C_{2\bar{x}y}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}11\bar{2}$ 3.4.11	$\{E, C_{2z}\mathcal{T}, \mathcal{T}, C_{2z}\}$	0	0
			$\mathrm{PL}_{(010)}$	p_cm11 11.7.56	$\{E, M_y\mathcal{T}, \mathcal{T}, M_y\}$	0	0
			$\mathrm{PL}_{(100)}$	p_cm11 11.7.56	$\{E, M_x\mathcal{T}, \mathcal{T}, M_x\}$	0	0
			$\mathrm{PL}_{(110)}$	$c_p2'11$ 10.5.49	$\{E, C_{2xy}\mathcal{T}, \mathcal{T}, C_{2xy}\}$	0	0
			$\mathrm{PL}_{(1\bar{1}0)}$	$c_p2'11$ 10.5.49	$\{E, C_{2\bar{x}y}\mathcal{T}, \mathcal{T}, C_{2\bar{x}y}\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0
118 59.7.416 $p_p^1\bar{4}^{-1}m^{-1}2$ SI p. 60	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|\bar{C}_{4z}\}$ $\{1\|\bar{C}_{4z}^{-1}\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$ $\{-1\|C_{2xy}\}$ $\{-1\|C_{2\bar{x}y}\}$ $\{-1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|\bar{C}_{4z}\}$ $\{-1\|\bar{C}_{4z}^{-1}\}$ $\{1\|M_y\}$ $\{1\|M_x\}$ $\{1\|C_{2xy}\}$ $\{1\|C_{2\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	p_cmm2 23.6.134	$\{E, C_{2z}\mathcal{T}, M_y, M_x\mathcal{T}, \mathcal{T}, C_{2z}, M_y\mathcal{T}, M_x\}$	0	0
			$\mathrm{AX}_{[010]}$	p_cmm2 23.6.134	$\{E, C_{2z}\mathcal{T}, M_y\mathcal{T}, M_x, \mathcal{T}, C_{2z}, M_y, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[110]}$	$c_p2'22'$ 22.7.128	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}, \mathcal{T}, C_{2z}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[1\bar{1}0]}$	$c_p2'22'$ 22.7.128	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}, \mathcal{T}, C_{2z}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}\}$	0	0
			$\mathrm{AX}_{[001]}$	p_p4m2 59.6.415	$\{E, C_{2z}, \bar{C}_{4z}, \bar{C}_{4z}^{-1}, M_y, M_x, C_{2xy}, C_{2\bar{x}y}, \mathcal{T}, C_{2z}\mathcal{T}, \bar{C}_{4z}\mathcal{T}, \bar{C}_{4z}^{-1}\mathcal{T}, M_y\mathcal{T}, M_x\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}11\bar{2}$ 3.4.11	$\{E, C_{2z}\mathcal{T}, \mathcal{T}, C_{2z}\}$	0	0
			$\mathrm{PL}_{(010)}$	p_cm11 11.7.56	$\{E, M_y, \mathcal{T}, M_y\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(100)}$	p_cm11 11.7.56	$\{E, M_x, \mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(110)}$	c_p211 10.4.48	$\{E, C_{2xy}, \mathcal{T}, C_{2xy}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(1\bar{1}0)}$	c_p211 10.4.48	$\{E, C_{2\bar{x}y}, \mathcal{T}, C_{2\bar{x}y}\mathcal{T}\}$	0	0
			GEN	$p_{2a}1$ 1.3.3	$\{E, \mathcal{T}\}$	0	0

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG (OG) symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
119 61.10.431 $p^1_p4/^{-1}m^{-1}m^{-1}m$ SI p. 60	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{4z}\}$ $\{1\|C^{-1}_{4z}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{1\|C_{2xy}\}$ $\{1\|C_{2\bar{x}y}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{-1\|\bar{C}_{4z}\}$ $\{-1\|\bar{C}^{-1}_{4z}\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$ $\{-1\|M_{xy}\}$ $\{-1\|M_{\bar{x}y}\}$ $\{-1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{4z}\}$ $\{-1\|C^{-1}_{4z}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{-1\|C_{2xy}\}$ $\{-1\|C_{2\bar{x}y}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{1\|\bar{C}_{4z}\}$ $\{1\|\bar{C}^{-1}_{4z}\}$ $\{1\|M_y\}$ $\{1\|M_x\}$ $\{1\|M_{xy}\}$ $\{1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	p_cmmm 37.11.240	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_z, M_y, M_x\mathcal{T}, \mathcal{T}, C_{2z}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	0
			$\mathrm{AX}_{[010]}$	p_cmmm 37.11.240	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z, M_y\mathcal{T}, M_x, \mathcal{T}, C_{2z}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[110]}$	$c_pmm'm$ 47.13.342	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z, M_{xy}\mathcal{T}, M_{\bar{x}y}, \mathcal{T}, C_{2z}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}, \mathcal{P}, M_z\mathcal{T}, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[1\bar{1}0]}$	$c_pmm'm$ 47.13.342	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}, \mathcal{P}\mathcal{T}, M_z, M_{xy}, M_{\bar{x}y}\mathcal{T}, \mathcal{T}, C_{2z}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_p4/m'mm$ 61.12.433	$\{E, C_{2z}, C_{4z}, C^{-1}_{4z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, \bar{C}_{4z}\mathcal{T}, \bar{C}^{-1}_{4z}\mathcal{T}, M_y, M_x, M_{xy}, M_{\bar{x}y}, \mathcal{T}, C_{2z}\mathcal{T}, C_{4z}\mathcal{T}, C^{-1}_{4z}\mathcal{T}, C_{2y}, C_{2x}, C_{2xy}, C_{2\bar{x}y}, \mathcal{P}, M_z, \bar{C}_{4z}, \bar{C}^{-1}_{4z}, M_y\mathcal{T}, M_x\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}112/m$ 6.6.26	$\{E, C_{2z}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z, \mathcal{T}, C_{2z}, \mathcal{P}, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	$p_c2/m11$ 14.8.73	$\{E, C_{2y}\mathcal{T}, \mathcal{P}\mathcal{T}, M_y, \mathcal{T}, C_{2y}, \mathcal{P}, M_y\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_c2/m11$ 14.8.73	$\{E, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_x, \mathcal{T}, C_{2x}, \mathcal{P}, M_x\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(110)}$	$c_p2'/m11$ 18.9.103	$\{E, C_{2xy}\mathcal{T}, \mathcal{P}\mathcal{T}, M_{xy}, \mathcal{T}, C_{2xy}, \mathcal{P}, M_{xy}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(1\bar{1}0)}$	$c_p2'/m11$ 18.9.103	$\{E, C_{2\bar{x}y}\mathcal{T}, \mathcal{P}\mathcal{T}, M_{\bar{x}y}, \mathcal{T}, C_{2\bar{x}y}, \mathcal{P}, M_{\bar{x}y}\mathcal{T}\}$	0	0
			GEN	$p_{2a}\bar{1}$ 2.4.7	$\{E, \mathcal{P}\mathcal{T}, \mathcal{T}, \mathcal{P}\}$	0	0

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG (OG) symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
120 (S2) 61.11.432 $p^1_p4/{}^1m^{-1}{}^1m^{-1}m$ SI p. 61	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{4z}\}$ $\{1\|C^{-1}_{4z}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{-1\|C_{2xy}\}$ $\{-1\|C_{2\bar{x}y}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{1\|\bar{C}_{4z}\}$ $\{1\|\bar{C}^{-1}_{4z}\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$ $\{-1\|M_{xy}\}$ $\{-1\|M_{\bar{x}y}\}$ $\{-1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{4z}\}$ $\{-1\|C^{-1}_{4z}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{1\|C_{2xy}\}$ $\{1\|C_{2\bar{x}y}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{-1\|\bar{C}_{4z}\}$ $\{-1\|\bar{C}^{-1}_{4z}\}$ $\{1\|M_y\}$ $\{1\|M_x\}$ $\{1\|M_{xy}\}$ $\{1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$p_cmm'm'$ 37.12.241	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}, \mathcal{T}, C_{2z}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_z, M_y\mathcal{T}, M_x\}$	0	0
			$\mathrm{AX}_{[010]}$	$p_cmm'm'$ 37.12.241	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}, M_z\mathcal{T}, M_y\mathcal{T}, M_x, \mathcal{T}, C_{2z}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z, M_y, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[110]}$	$c_pmm'm'$ 47.11.340	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}, \mathcal{P}, M_z\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}, \mathcal{T}, C_{2z}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[1\bar{1}0]}$	$c_pmm'm'$ 47.11.340	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_{xy}, M_{\bar{x}y}\mathcal{T}, \mathcal{T}, C_{2z}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}, \mathcal{P}\mathcal{T}, M_z, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	0	0
			$\mathrm{AX}_{[001]}$	p_p4/mmm 61.13.434	$\{E, C_{2z}, C_{4z}, C^{-1}_{4z}, C_{2y}, C_{2x}, C_{2xy}, C_{2\bar{x}y}, \mathcal{P}, M_z, \bar{C}_{4z}, \bar{C}^{-1}_{4z}, M_y, M_x, M_{xy}, M_{\bar{x}y}, \mathcal{T}, C_{2z}\mathcal{T}, C_{4z}\mathcal{T}, C^{-1}_{4z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, \bar{C}_{4z}\mathcal{T}, \bar{C}^{-1}_{4z}\mathcal{T}, M_y\mathcal{T}, M_x\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}112'/m'$ 6.7.27	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, \mathcal{T}, C_{2z}, \mathcal{P}\mathcal{T}, M_z\}$	0	0
			$\mathrm{PL}_{(010)}$	$p_c2/m11$ 14.8.73	$\{E, C_{2y}, \mathcal{P}, M_y, \mathcal{T}, C_{2y}\mathcal{T}, \mathcal{P}\mathcal{T}, M_y\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_c2/m11$ 14.8.73	$\{E, C_{2x}, \mathcal{P}, M_x, \mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(110)}$	$c_p2/m11$ 18.6.100	$\{E, C_{2xy}, \mathcal{P}, M_{xy}, \mathcal{T}, C_{2xy}\mathcal{T}, \mathcal{P}\mathcal{T}, M_{xy}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(1\bar{1}0)}$	$c_p2/m11$ 18.6.100	$\{E, C_{2\bar{x}y}, \mathcal{P}, M_{\bar{x}y}, \mathcal{T}, C_{2\bar{x}y}\mathcal{T}, \mathcal{P}\mathcal{T}, M_{\bar{x}y}\mathcal{T}\}$	0	0
			GEN	$p_{2a}\bar{1}$ 2.4.7	$\{E, \mathcal{P}, \mathcal{T}, \mathcal{P}\mathcal{T}\}$	0	0

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG (OG) symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
121 61.12.433 $p^1_p4/^{-1}m^1m^1m$ SI p. 62	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{4z}\}$ $\{1\|C^{-1}_{4z}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{-1\|C_{2xy}\}$ $\{-1\|C_{2\bar{x}y}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{-1\|\bar{C}_{4z}\}$ $\{-1\|\bar{C}^{-1}_{4z}\}$ $\{1\|M_y\}$ $\{1\|M_x\}$ $\{1\|M_{xy}\}$ $\{1\|M_{\bar{x}y}\}$ $\{-1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{4z}\}$ $\{-1\|C^{-1}_{4z}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{1\|C_{2xy}\}$ $\{1\|C_{2\bar{x}y}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{1\|\bar{C}_{4z}\}$ $\{1\|\bar{C}^{-1}_{4z}\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$ $\{-1\|M_{xy}\}$ $\{-1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	p_cmmm 37.11.240	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{PT}, M_z, M_y\mathcal{T}, M_x, \mathcal{T}, C_{2z}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	p_cmmm 37.11.240	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{PT}, M_z, M_y, M_x\mathcal{T}, \mathcal{T}, C_{2z}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	0
			$\mathrm{AX}_{[110]}$	$c_pmm'm$ 47.13.342	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}, \mathcal{PT}, M_z, M_{xy}, M_{\bar{x}y}\mathcal{T}, \mathcal{T}, C_{2z}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	0	0
			$\mathrm{AX}_{[1\bar{1}0]}$	$c_pmm'm$ 47.13.342	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}, \mathcal{PT}, M_z, M_{xy}\mathcal{T}, M_{\bar{x}y}, \mathcal{T}, C_{2z}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}, \mathcal{P}, M_z\mathcal{T}, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_p4/m'm'm'$ 61.10.431	$\{E, C_{2z}, C_{4z}, C^{-1}_{4z}, C_{2y}, C_{2x}, C_{2xy}, C_{2\bar{x}y}, \mathcal{PT}, M_z\mathcal{T}, \bar{C}_{4z}\mathcal{T}, \bar{C}^{-1}_{4z}\mathcal{T}, M_y\mathcal{T}, M_x\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}\mathcal{T}, \mathcal{T}, C_{2z}\mathcal{T}, C_{4z}\mathcal{T}, C^{-1}_{4z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}\mathcal{T}, \mathcal{P}, M_z, \bar{C}_{4z}, \bar{C}^{-1}_{4z}, M_y, M_x, M_{xy}, M_{\bar{x}y}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}112/m$ 6.6.26	$\{E, C_{2z}\mathcal{T}, \mathcal{PT}, M_z, \mathcal{T}, C_{2z}, \mathcal{P}, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	$p_c2/m11$ 14.8.73	$\{E, C_{2y}, \mathcal{PT}, M_y\mathcal{T}, \mathcal{T}, C_{2y}\mathcal{T}, \mathcal{P}, M_y\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_c2/m11$ 14.8.73	$\{E, C_{2x}, \mathcal{PT}, M_x\mathcal{T}, \mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}, M_x\}$	0	0
			$\mathrm{PL}_{(110)}$	$c_p2/m'11$ 18.7.101	$\{E, C_{2xy}, \mathcal{PT}, M_{xy}\mathcal{T}, \mathcal{T}, C_{2xy}\mathcal{T}, \mathcal{P}, M_{xy}\}$	0	0
			$\mathrm{PL}_{(1\bar{1}0)}$	$c_p2/m'11$ 18.7.101	$\{E, C_{2\bar{x}y}, \mathcal{PT}, M_{\bar{x}y}\mathcal{T}, \mathcal{T}, C_{2\bar{x}y}\mathcal{T}, \mathcal{P}, M_{\bar{x}y}\}$	0	0
			GEN	$p_{2a}\bar{1}$ 2.4.7	$\{E, \mathcal{PT}, \mathcal{T}, \mathcal{P}\}$	0	0

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG (OG) symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
122 61.13.434 $p^1_p4/{}^1m{}^1m{}^1m$ SI p. 62	$T\tau$ AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{4z}\}$ $\{1\|C^{-1}_{4z}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{1\|C_{2xy}\}$ $\{1\|C_{2\bar{x}y}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{1\|\bar{C}_{4z}\}$ $\{1\|\bar{C}^{-1}_{4z}\}$ $\{1\|M_y\}$ $\{1\|M_x\}$ $\{1\|M_{xy}\}$ $\{1\|M_{\bar{x}y}\}$ $\{-1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{4z}\}$ $\{-1\|C^{-1}_{4z}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{-1\|C_{2xy}\}$ $\{-1\|C_{2\bar{x}y}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{-1\|\bar{C}_{4z}\}$ $\{-1\|\bar{C}^{-1}_{4z}\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$ $\{-1\|M_{xy}\}$ $\{-1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$p_cmm'm'$ 37.12.241	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}, M_z\mathcal{T}, M_y\mathcal{T}, M_x, \mathcal{T}, C_{2z}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z, M_y, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	$p_cmm'm'$ 37.12.241	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}, \mathcal{T}, C_{2z}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_z, M_y\mathcal{T}, M_x\}$	0	0
			$\mathrm{AX}_{[110]}$	$c_pmm'm'$ 47.11.340	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_{xy}, M_{\bar{x}y}\mathcal{T}, \mathcal{T}, C_{2z}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}, \mathcal{P}\mathcal{T}, M_z, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	0	0
			$\mathrm{AX}_{[1\bar{1}0]}$	$c_pmm'm'$ 47.11.340	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}, \mathcal{P}, M_z\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}, \mathcal{T}, C_{2z}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$p_p4/mm'm'$ 61.11.432	$\{E, C_{2z}, C_{4z}, C^{-1}_{4z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}\mathcal{T}, \mathcal{P}, M_z, \bar{C}_{4z}, \bar{C}^{-1}_{4z}, M_y\mathcal{T}, M_x\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}\mathcal{T}, \mathcal{T}, C_{2z}\mathcal{T}, C_{4z}\mathcal{T}, C^{-1}_{4z}\mathcal{T}, C_{2y}, C_{2x}, C_{2xy}, C_{2\bar{x}y}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, \bar{C}_{4z}\mathcal{T}, \bar{C}^{-1}_{4z}\mathcal{T}, M_y, M_x, M_{xy}, M_{\bar{x}y}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p_{2a}112'/m'$ 6.7.27	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, \mathcal{T}, C_{2z}, \mathcal{P}\mathcal{T}, M_z\}$	0	0
			$\mathrm{PL}_{(010)}$	$p_c2/m11$ 14.8.73	$\{E, C_{2y}\mathcal{T}, \mathcal{P}, M_y\mathcal{T}, \mathcal{T}, C_{2y}, \mathcal{P}\mathcal{T}, M_y\}$	0	0
			$\mathrm{PL}_{(100)}$	$p_c2/m11$ 14.8.73	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}, \mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_x\}$	0	0
			$\mathrm{PL}_{(110)}$	$c_p2'/m'11$ 18.8.102	$\{E, C_{2xy}\mathcal{T}, \mathcal{P}, M_{xy}\mathcal{T}, \mathcal{T}, C_{2xy}, \mathcal{P}\mathcal{T}, M_{xy}\}$	0	0
			$\mathrm{PL}_{(1\bar{1}0)}$	$c_p2'/m'11$ 18.8.102	$\{E, C_{2\bar{x}y}\mathcal{T}, \mathcal{P}, M_{\bar{x}y}\mathcal{T}, \mathcal{T}, C_{2\bar{x}y}, \mathcal{P}\mathcal{T}, M_{\bar{x}y}\}$	0	0
			GEN	$p_{2a}\bar{1}$ 2.4.7	$\{E, \mathcal{P}, \mathcal{T}, \mathcal{P}\mathcal{T}\}$	0	0

Appendix A. Why every layer response vanishes

The anti-translation $g_\tau = \{\mathcal{T} \mid \boldsymbol{\tau}\}$ has spatial point part $R = E$, time-reversal parity $q = -1$ and $\tau_z = 0$. It preserves the identity of each physical layer. For a homogeneous electric field and the response averaged over the full magnetic in-plane cell,

$$\chi^{\ell}_{abc} \xrightarrow{g_\tau} -\chi^{\ell}_{abc}, \qquad \ell = t, b.$$

Neumann invariance therefore imposes $\chi^{\ell}_{abc} = -\chi^{\ell}_{abc}$ and hence $\chi^{\ell}_{abc} = 0$ for the time-reversal-odd intrinsic Hall part. In the four-component representation used throughout the tables,

$$(A^t, B^t, A^b, B^b) = (0, 0, 0, 0), \qquad \dim \mathcal{V}_{\text{allowed}} = 0.$$

The constraint holds for every Néel direction because the spatial point part of the spin-reversing translation is the identity. Rotating the common spin axis does not remove g_τ from the SOC subgroup. The conclusion concerns the physical-layer projection used in the table; a projection onto a smaller sublattice exchanged by $\boldsymbol{\tau}$ need not obey the same local-zero statement.

Magnetic translation lattices in the source OG coordinates

The columns below span the unitary translation lattice L_+ in the source family-cell basis. Listed anti-translations are representatives modulo L_+. The pair q, R alone is insufficient to identify the full magnetic layer group.

OG lattice prefix	Basis of L_+	Anti-translation representative
p_{2a}	$(2\mathbf{a}, \mathbf{b})$	$\mathbf{a}$
p_{2b}	$(\mathbf{a}, 2\mathbf{b})$	$\mathbf{b}$
p_c	$(\mathbf{a}+\mathbf{b}, -\mathbf{a}+\mathbf{b})$	$\mathbf{b}$
p_p	$(\mathbf{a}+\mathbf{b}, -\mathbf{a}+\mathbf{b})$	$\mathbf{a}$
c_p	$(\mathbf{a}, \mathbf{b})$	$(\mathbf{a}+\mathbf{b})/2$

If P contains these basis vectors as columns, affine operations are normalized as $R_m = P^{-1}RP$ and $t_m = P^{-1}t$. Translations are reduced modulo integer vectors only after this transformation. This is essential: reducing an anti-translation $(1, 0, 0)$ modulo the source family cell would incorrectly turn it into a pure time reversal.

Every reported MLG has an explicit affine witness preserving the layer normal. The data package gives the source operations, normalized magnetic-cell basis, SOC subgroup operations, full allowed tensor space and the basis/origin transformation. Group labels and MLG primes are in the reference OG setting, while the Operation in MLG column remains in the source Cartesian setting.